

ABSTRACT

Distillery industries generate large volumes of highly polluted wastewater known as Raw Spent Wash (RSW), characterized by high organic load, dark colour, acidity, and recalcitrant compounds such as melanoidins and phenolics. This study investigates the effect of hydrodynamic cavitation (HC) pretreatment on Raw Spent Wash (RSW) and Treated Spent Wash (TSW) to enhance biodegradability and methane production potential. Initially, the RSW showed a COD of 128000 mg/L, TDS of 106900 mg/L, and VS/TS ratio of 72.06%, indicating a high biodegradable organic fraction suitable for anaerobic digestion. In contrast, TSW exhibited a COD of 40000 mg/L and VS/TS ratio of 43.03%, reflecting partial stabilization after prior treatment.

Hydrodynamic cavitation experiments were carried out using a pilot-scale venturi-based HC setup operated at a flow rate of 3 m³/hr with four sequential passes. Cavitation treatment caused partial breakdown of complex organic compounds and improved biodegradability of TSW. After HC treatment, COD of TSW reduced from 40000 mg/L to 28000 mg/L, while pH increased from 8.2 to 8.5. Biochemical Methane Potential (BMP) analysis was conducted over a digestion period of 20 days using an AMPTS system. Untreated RSW showed the highest methane yield of approximately 410 NmL CH₄/g VS, while RSW after HC treatment showed a slightly lower BMP value of 400 NmL CH₄/g VS, indicating negligible improvement. However, HC pretreatment significantly enhanced the methane yield of TSW from 15.6 to 113.6 NmL CH₄/g VS, corresponding to nearly a 7.2-fold increase in methane potential.

The results demonstrate that hydrodynamic cavitation is highly effective for improving the biodegradability of treated distillery effluent by converting non-biodegradable COD into biodegradable fractions. Therefore, HC can be considered an efficient and sustainable pretreatment technology for enhancing biomethane recovery from treated spent wash and improving distillery wastewater management.

Chapter 1
INTRODUCTION

1. INTRODUCTION

1.1 Background on Renewable Energy and Biogas

The growing global energy demand, coupled with the rapid depletion of conventional fossil fuel resources, has intensified the search for renewable and environmentally sustainable alternatives. Among the various emerging technologies, anaerobic digestion has gained prominence as a promising renewable energy solution capable of converting organic waste into biogas a clean, carbon-neutral fuel composed mainly of methane and carbon dioxide. Biogas production not only provides an alternative energy source but also supports waste-to-energy initiatives, making it particularly attractive for industries generating high-strength organic effluents. In recent years, there has been a significant rise in interest toward biogas generation from industrial waste streams, owing to its dual benefits in energy recovery and wastewater stabilization, thereby contributing to a circular and sustainable bio-economy [1].

1.2 Molasses-based Distillery Industry and Spent Wash Generation

The distillery industry plays a significant role in the global bioethanol and beverage sectors, particularly in countries like India, Brazil, and the United States. However, the production of ethanol through the fermentation of molasses or other sugar-rich substrates generates a highly polluting liquid effluent known as spent wash or distillery wastewater. Among all types of industrial effluents, raw spent wash is considered one of the most complex and difficult-to-treat wastewaters, due to its high organic load, dark coloration, low pH, and toxic constituents. On average, 1 liter of ethanol production generates 10–15 liters of spent wash, highlighting the magnitude of waste generation in this sector [2].

Raw spent wash is the residual liquid left after the distillation of fermented molasses or grain mash in ethanol production units. In a typical distillery, sugarcane molasses is diluted with water, inoculated with yeast for fermentation, and then distilled to extract alcohol. The remaining residue from the distillation column is the spent wash, which is discharged as a dark brown liquid rich in organic and inorganic materials. Depending on the feedstock (molasses, cane juice, grains, or sugar beet), the characteristics of spent wash may vary, but its overall pollutant load remains extremely high [3].

Raw spent wash is characterized by extremely high pollution potential. It is acidic (pH 3.5–5.0) and contains a very high Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), indicating a large quantity of organic matter. The COD of raw spent wash typically ranges between 80,000–120,000 mg/L, while the BOD ranges from 40,000–60,000 mg/L [4]. It also contains a high amount of Total Dissolved Solids (TDS) (50,000–80,000 mg/L), Total Suspended Solids (TSS) (5,000–10,000 mg/L), and volatile organic compounds, making it unsuitable for direct discharge into water bodies. The characteristics of RSW are also shown in Table 1.2 below. The dark brown color of raw spent wash results from the presence of melanoidins, which are complex, high-molecular-weight polymers formed through Maillard reactions between amino acids and reducing sugars during fermentation and distillation. These compounds are non-biodegradable and resistant to conventional biological degradation, contributing to color persistence even after dilution. In addition to melanoidins, raw spent wash contains phenolic compounds, furfurals, sulfates, chlorides, heavy metals (Fe, Cu, Mn, Zn), and nutrients (N, P, K).

Table 1.2 Characteristics of Raw Spent Wash [4]

Sr. No.	Parameter	Range
1	pH	3.5-5
2	Color	Dark brown to black
3	COD	80,000-120,000 mg/L
4	BOD	40,000-60,000 mg/L
5	Total Solids	70,000-120,000 mg/L
6	TDS	50,000-80,000 mg/L
7	TSS	5000-10,000 mg/L
8	Sulfates	2000-5000 mg/L
9	Chlorides	2000-4000 mg/L
10	Nitrogen	1000-2500 mg/L
11	Potassium	1000-2000 mg/L

Raw spent wash poses severe environmental threats when discharged untreated into the environment. Its high organic load leads to rapid depletion of dissolved oxygen (DO) in water bodies, severely affecting aquatic life. The color and turbidity prevent light penetration, inhibiting photosynthetic activity in aquatic ecosystems. The presence of phenolic compounds and melanoidins imparts toxicity to microorganisms and plants,

resulting in reduced soil fertility and inhibition of seed germination when used for irrigation without treatment.

When disposed of on land, the high TDS and salinity cause soil hardening and disturb the ionic balance, making the soil infertile over time. Leaching of these pollutants can contaminate groundwater with toxic ions and heavy metals. Moreover, the anaerobic decomposition of spent wash generates foul odor and greenhouse gases (CH_4 , CO_2 , H_2S), contributing to air pollution and climate change [3].

The treatment of raw spent wash is technically challenging due to its high strength, complex composition, and poor biodegradability. Conventional biological methods such as anaerobic digestion can achieve partial COD and BOD reduction but fail to remove color and recalcitrant organics completely. Physicochemical methods like coagulation, adsorption, and membrane filtration are effective for partial decolorization but are costly and generate secondary sludge. To overcome these limitations, researchers have explored Advanced Oxidation Processes (AOPs) such as ozonation, photocatalysis, Fenton oxidation, and hydrodynamic cavitation, which generate highly reactive hydroxyl radicals ($\bullet\text{OH}$) capable of breaking down complex organic molecules. Among these, hydrodynamic cavitation and ozone-based AOPs have shown promising results in degrading melanoidins and reducing color, COD, and toxicity in spent wash [5].

1.3 Treated Spent Wash

Treated spent wash is the effluent obtained after subjecting raw distillery spent wash to suitable treatment processes for reducing its pollutant load and improving its environmental compatibility. Raw spent wash generated from molasses-based distilleries is highly acidic, dark brown in colour, and contains high concentrations of organic matter, dissolved solids, suspended solids, and recalcitrant compounds such as melanoidins. Therefore, direct discharge of raw spent wash can cause serious environmental problems, including soil degradation, water pollution, and depletion of dissolved oxygen in receiving water bodies.

To minimize these impacts, spent wash is commonly treated by anaerobic digestion, in which microorganisms degrade biodegradable organic matter in the absence of oxygen and generate biogas rich in methane. The effluent obtained after anaerobic digestion is known as treated spent wash (TSW). Compared to raw spent wash, TSW generally

shows lower COD, BOD, and volatile solids, indicating that a large portion of biodegradable organic matter has already been consumed during biogas production. However, TSW still contains high total dissolved solids, dark color, and residual non-biodegradable organic compounds, mainly melanoidins and other recalcitrant substances. Due to the presence of these refractory compounds, further treatment or advanced oxidation processes are often required to improve the biodegradability and achieve additional pollutant removal. The characteristics of RSW are presented in Table 1.3 below.

Treated spent wash has gained importance due to its potential for reuse and resource recovery. It can be utilized for irrigation after proper dilution, composting, ferti-irrigation, or further polishing treatments. In research studies, treated spent wash is often investigated to evaluate the effectiveness of advanced treatment techniques such as hydrodynamic cavitation, ozonation, or photocatalysis for additional reduction of pollutants. Thus, treated spent wash represents a partially stabilized wastewater stream that offers opportunities for sustainable management and circular utilization in the distillery industry.

Table 1.3 Characteristics of Treated Spent Wash

Sr. No.	Parameter	Range
1	pH	3.5-5
2	Color	Dark brown to black
3	COD	80,000-120,000 mg/L
4	BOD	40,000-60,000 mg/L
5	Total Solids	70,000-120,000 mg/L
6	TDS	50,000-80,000 mg/L
7	TSS	5000-10,000 mg/L
8	Sulfates	2000-5000 mg/L
9	Chlorides	2000-4000 mg/L
10	Nitrogen	1000-2500 mg/L
11	Potassium	1000-2000 mg/L

1.4 Manufacturing of Raw Spent Wash from Cane molasses

1.4.1 Sugarcane Molasses Production

In order to concentrate and crystallize sucrose, the juice from sugarcane is first crushed in mills, clarified, and then heated under vacuum. Centrifugation is used to extract the sugar crystals, leaving behind molasses, a thick, dark-brown liquid. This molasses is a

good substrate for additional fermentation processes because it preserves organic chemicals, minerals, and non-crystallizable sugars (fructose and glucose) [6].

1.4.2 Fermentation of Molasses

To encourage microbial development, molasses, which is high in fermentable sugars, is diluted with water to the appropriate concentration and then enhanced with nutrients like phosphates and nitrogen. After that, the medium is supplemented with yeast, most frequently *Saccharomyces cerevisiae*, which anaerobically breaks down carbohydrates into carbon dioxide and ethanol. After a normal 24- to 30-hour procedure, an alcohol-rich fermented wash with 7–10% ethanol is produced, which is then delivered for distillation [7].

1.4.3 Distillation of Fermented Wash

After yeast fermentation, the alcohol-rich fermented wash is put into distillation columns, where it is boiled to separate ethanol from water and other contaminants according to their respective boiling points. Depending on the method employed, ethanol vapors are collected and condensed at a purity of 95–99%. Raw wasted wash, the leftover dark brown, high-strength liquid effluent that is released after alcohol recovery, has a high organic load and presents significant treatment issues [4].

1.4.4 Characteristics of Raw Spent Wash

Raw spent wash is a dark brown, acidic wastewater that is created after alcohol is recovered during molasses distillation. It has a very high organic load, as evidenced by its high chemical oxygen demand (COD) and biological oxygen demand (BOD). The effluent is rich in potassium, sulfates, and chlorides and contains melanoidin pigments, which give it its color. Because of its potent odor and enormous production nearly 8 to 15 liters for every liter of ethanol produced its management poses a serious environmental risk [8].

1.5 Environmental Challenges Associated of Raw Spent Wash

The environmental impact of spent wash disposal is severe when untreated. Its dark colour and turbidity obstruct light penetration in aquatic environments, reducing photosynthetic oxygenation and harming aquatic biodiversity. Its acidic pH and high dissolved solids contribute to soil hardening, altered nutrient exchange, and long-term

loss of agricultural productivity. Toxic compounds such as melanoidins, phenols, furfurals, and sulphates impart phytotoxicity, inhibit seed germination, and suppress microbial activity crucial for soil fertility. Further, uncontrolled anaerobic degradation leads to the emission of CO₂, CH₄, and H₂S, contributing to air pollution and greenhouse gas accumulation. Leaching and percolation of spent wash contaminate groundwater with salts and toxic ions, posing risks to human and ecological health. These challenges have necessitated stricter regulatory mandates and exploration of sustainable treatment solutions [3].

1.6 Anaerobic Digestion of Spent Wash

Distillery spent wash is considered a highly attractive substrate for biomethanation due to its extremely high Chemical Oxygen Demand (COD), indicating a large fraction of biodegradable organic matter that can be converted into methane through anaerobic digestion. The theoretical biogas generation potential increases proportionally with COD concentration, making spent wash a valuable resource for renewable energy production while simultaneously reducing the pollution load. However, despite this high energy potential, the anaerobic digestion of raw spent wash faces several critical limitations. The effluent contains recalcitrant and structurally complex organic compounds, such as melanoidins, phenolics, furfurals, and lignin-derived polymers, which degrade very slowly and hinder the hydrolysis stage of digestion. Additionally, many of these compounds exert inhibitory or toxic effects on methanogenic microorganisms, disrupting cell integrity, altering metabolic pathways, and suppressing methanogenesis due to factors like high salinity, phenolic toxicity, and sulfide formation. As a result, the methane yield achieved from untreated spent wash is typically much lower than the theoretical potential, and digestion becomes unstable, requiring long retention times and careful operational control. Therefore, pre-treatment of spent wash is essential to break down complex organics, reduce inhibitory components, enhance solubilization, and improve microbial accessibility, ultimately boosting methane productivity and enabling more efficient biogas generation [9].

1.7 Importance of Pre-Treatment Methods

Pre-treatment plays a crucial role in enhancing the efficiency of anaerobic digestion for distillery spent wash and other high-strength industrial effluents. Because raw spent wash contains a significant fraction of complex, recalcitrant organic compounds—

melanoidins, lignin-derived polymers, phenolics, and long-chain molecules—its direct biodegradation under anaerobic conditions is slow and inefficient. Pre-treatment methods aim to break down these complex structures, improve solubilization of organic matter, reduce microbial inhibitors, and enhance the accessibility of substrates to methanogenic microorganisms, thereby improving biodegradability and methane yield [1].

1.7.1 Overview of Pre-Treatment Techniques

Pre-treatment approaches for improving anaerobic digestion are broadly categorized into physical, chemical, thermal, and biological methods, with hydrodynamic cavitation (HC) emerging as one of the most promising physical techniques [10]: Among physical pre-treatments, HC stands out because the rapid formation and violent collapse of microbubbles generate intense shear forces, shock waves, and localized hotspots that effectively rupture cell walls, reduce particle size, and markedly increase the solubilization of organic matter. Compared to other physical methods such as ultrasonication, milling, or mechanical stirring, HC offers higher energy efficiency and better scalability for industrial wastewater applications. Chemical methods including alkaline reagents (NaOH, Ca(OH)_2), acid hydrolysis, ozonation, and H_2O_2 oxidation aim to cleave complex polymers and enhance solubility, though they often involve higher chemical consumption and sludge generation. Thermal treatments like steam explosion, autoclaving, and high-temperature hydrolysis improve solubilization through structural weakening under elevated temperatures but may incur high energy costs. Biological pre-treatments, involving enzymes or specialized bacteria, convert large organic molecules into simpler intermediates via enzymatic pathways, though they typically proceed at slower rates. Overall, hydrodynamic cavitation provides a highly effective and energy-efficient means of converting non-biodegradable or slowly degradable components into readily digestible forms, making it a superior pre-treatment option for enhancing anaerobic digestion performance.

1.7.2 Role of Pre-Treatment in Improving Biodegradability and Methane Yield

Pre-treatment is vital for improving the anaerobic digestion of complex and high-strength effluents such as distillery spent wash, as it enhances overall biodegradability and increases methane output. In this study, hydrodynamic cavitation (HC) is used as the pre-treatment method, wherein the collapse of microbubbles generates intense shear

forces that break down resistant organic structures. This action increases the solubilization of organic matter, raises soluble COD levels, and improves substrate bioavailability thereby accelerating the hydrolysis phase, which is often the rate-limiting step in anaerobic digestion. HC also modifies or reduces inhibitory components such as phenolics, sulfates, and melanoidin complexes, minimizing their toxic effects on methanogens and stabilizing reactor performance. Additionally, the method enhances the Biodegradability Index (BOD/COD ratio), facilitating more efficient conversion of complex organics into methane. By reducing volatile fatty acid accumulation and enabling operation at higher organic loading rates, hydrodynamic cavitation ensures smoother digestion with fewer process disturbances. Ultimately, applying HC as the pre-treatment method leads to improved methane yield, faster biogas production, and reduced hydraulic retention time, strengthening the economic and industrial feasibility of treating distillery spent wash [10].

1.8 Hydrodynamic Cavitation

Cavitation occurs when the local pressure falls below the liquid's vapor pressure, causing vapor-filled cavities (bubbles) to form, grow, then violently collapse. When high-velocity flows pass through constrictions like orifices, venturi tubes, pump impellers, or when liquids are exposed to ultrasonic waves, this phenomenon usually happens. Three steps make up cavitation: microbubbles are nucleated at impurity or dissolved gas sites, develop as they enter low-pressure areas, and quickly collapse as they enter higher-pressure areas. Extreme localized conditions, such as temperatures exceeding 5000 K, pressures above 1000 atm, high shear, and shock waves, are created when these bubbles implode and can have either positive or negative impacts. In pumps, turbines, and propellers, unchecked cavitation causes erosion, vibration, noise, and energy loss [11]. However, there are many uses for controlled cavitation, especially hydrodynamic and acoustic cavitation, including food processing, wastewater treatment, disinfection, sludge disintegration, emulsification, the extraction of bioactive compounds, and medical procedures like lithotripsy. Water molecules disintegrate during collapse to produce very reactive hydroxyl radicals ($\bullet\text{OH}$), which facilitate sophisticated oxidation processes that break down organic contaminants. The cavitation number (σ) is a measure of cavitation intensity; a lower value denotes more vigorous cavitation. Its potential as an environmentally friendly and energy-efficient solution for chemical processing and environmental cleanup is highlighted by recent studies [12].

Its synergistic application with catalysts (e.g., TiO_2 , ZnO) and oxidants (H_2O_2 , O_3) has shown remarkable improvements in the reduction of Chemical Oxygen Demand (COD), Total Organic Carbon (TOC), and color from industrial effluents such as textile dye wastewater, distillery spent wash, and pharmaceutical waste. Overall, hydrodynamic cavitation serves as an eco-friendly, energy-efficient, and scalable technique that enhances degradation kinetics and supports sustainable wastewater management [13].

When a liquid is exposed to high-frequency sound waves, usually between 20 kHz and 1 MHz, it undergoes acoustic cavitation, often referred to as ultrasonic cavitation, which produces alternating cycles of compression and rarefaction. These cycles produce explosive microbubble growth and collapse, releasing a localized high temperature (about 5000 K) and pressure (around 1000 atm). Hydroxyl radicals ($\bullet\text{OH}$) and other reactive species that can oxidize and break down organic contaminants are produced during this collapse. Due to its high energy requirement, acoustic cavitation is usually used at a laboratory or pilot size and is primarily regulated by ultrasonic frequency and intensity. It finds applications in sonochemical oxidation, disinfection, emulsification, nanomaterial production, and degradation of colors and medicines in wastewater treatment [14].

Hydrodynamic cavitation occurs when a liquid passes through a constriction such as a venturi tube or orifice plate, resulting in a localized reduction in pressure below the liquid's vapor pressure. Similar to acoustic cavitation, this results in the formation of vapor cavities that collapse downstream as the pressure recovers, producing high temperatures, strong local energy, and reactive radicals. Hydrodynamic cavitation is more energy-efficient and scalable for large-scale wastewater treatment than ultrasonic cavitation because it is caused by fluid dynamics rather than external sound energy. It is used to improve chemical and biological oxidation processes, decolorize textile effluents, break down sludge, and degrade organic contaminants. Hydrodynamic cavitation is often combined with oxidants (H_2O_2 , O_3) or catalysts (TiO_2 , ZnO) in hybrid advanced oxidation processes to improve treatment performance [15].

1.9 Types of Cavitation

Cavitation is a phenomenon where vapor bubbles form and collapse due to rapid pressure changes in a liquid, generating intense temperatures, pressures, and shear forces. These effects make cavitation highly useful in wastewater treatment and other

process-intensification applications, making it important to understand its different types.

Cavitation is primarily classified into tension-based and energy deposition types. Tension cavitation methods, such as hydraulic (hydrodynamic) and acoustic cavitation, serve as key advanced processes for wastewater remediation. The implosive collapse of cavitation bubbles produces extreme localized temperatures, pressures, and shear forces that drive pollutant breakdown, improve mass transfer rates, and generate reactive radicals capable of mineralizing recalcitrant organic matter. Hydraulic cavitation demonstrates superior energy efficiency and scalability for industrial applications, whereas acoustic cavitation provides finer control and elevated degradation rates, rendering both approaches effective for boosting wastewater biodegradability and treatment efficiency [8].

Energy-deposition cavitation methods, such as optic and particle cavitation, play key roles in advanced material processing and remediation processes. Optic cavitation arises from high-intensity laser irradiation, which induces localized optical breakdown and vapor bubble formation, enabling targeted energy delivery for applications in materials fabrication, biological treatments, and pollutant elimination. Particle cavitation results from interactions between solid particles and fluid motion, which amplify cavitation intensity to improve oxidation reactions, enhance mixing, and accelerate the breakdown of organic contaminants. Integrating these cavitation types boosts overall process performance and presents promising opportunities for industrial-scale and environmental remediation technologies [9]. In this study, we are mainly focusing on Hydraulic cavitation as it is used for the treatment of wastewater.

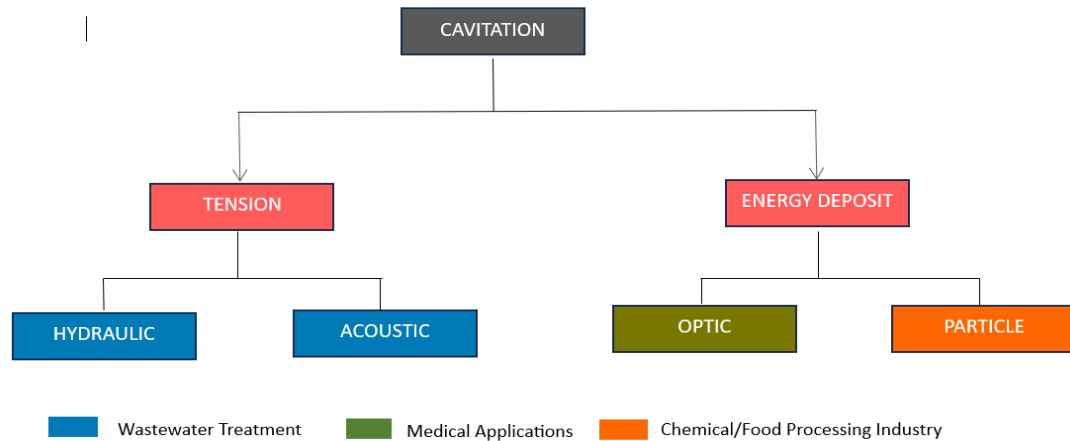


Figure 1.9 Types of Cavitation

1.9.1 Acoustic Cavitation

When high-intensity ultrasonic waves travel through a liquid, they produce alternating cycles of compression and rarefaction, which cause small bubbles to develop, grow, and violently collapse. This phenomenon is known as acoustic cavitation. Usually starting around dissolved gasses or other contaminants in the liquid, these bubbles grow throughout rarefaction cycles before popping under compression. Extreme localized circumstances are created when these bubbles implode, including shock waves, micro-jets, chemically active free radicals like hydroxyl radicals, and temperatures as high as 5000 K and pressures as high as 1000 atm [16]. Acoustic cavitation's strong physical and chemical properties make it extremely beneficial in a variety of industries, including the food and beverage industry for microbial inactivation and homogenization, the energy and environmental sectors for biomass pretreatment and improving biogas production, the medical field for therapeutic ultrasound, drug delivery, and lithotripsy, and the water and wastewater treatment industry for breaking down persistent pollutants. Acoustic cavitation can be advantageous when managed, however because of its extremely damaging localized stresses, it can also harm equipment [17].

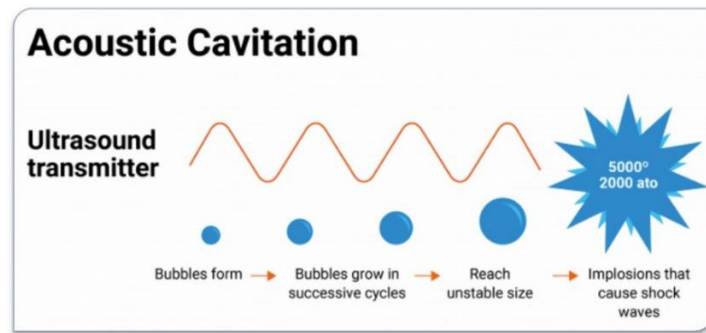


Figure 1.9.1 Acoustic Cavitation [18]

1.9.2 Hydraulic Cavitation

The process known as hydraulic cavitation, or hydrodynamic cavitation, happens when a liquid passes through an area of abrupt pressure drop, vaporizing and producing tiny vapor bubbles. These bubbles burst violently when they reach an area of higher pressure, unleashing powerful energy in the form of micro-jets, shock waves, and concentrated high pressures and temperatures [19] or in other words Cavitation is a phenomena where vapor bubbles form, grow, and then abruptly collapse as a result of rapid pressure changes in a liquid, typically brought on by high-velocity flow, obstacles, valves, or pumps. Controlled cavitation has been used more and more in industrial and environmental processes, but uncontrolled cavitation is undesirable in hydraulic gear because it results in erosion, noise, vibration, and loss of efficiency. Research has demonstrated its efficacy in pretreatment of biomass to increase the yields of biogas production, emulsification and microbial cell disruption for biotechnological applications, and wastewater treatment for lowering organic load and heavy metals. For example, it has been shown that hydrodynamic cavitation pretreatment greatly boosts the amount of methane produced from lignocellulosic feedstocks, such as wheat straw, increasing the efficiency of producing renewable energy.

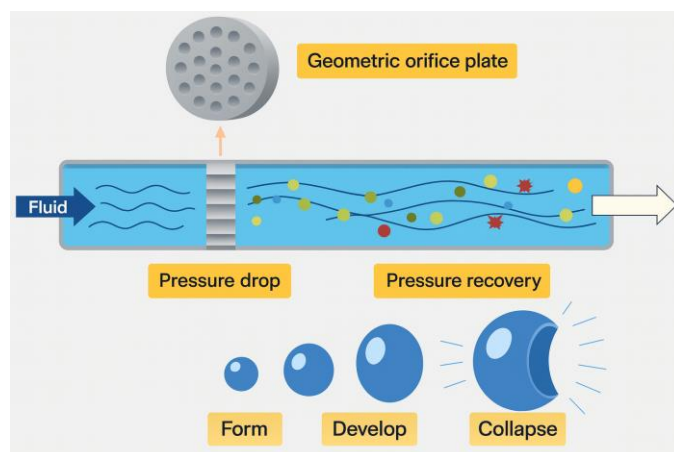


Figure 1.9.2 Hydraulic or Hydrodynamic Cavitation [20]

1.10 Application of Hydrodynamic Cavitation (HC) to Distillery Spent Wash

Hydrodynamic cavitation (HC) has emerged as a highly promising pretreatment method for molasses-based distillery spent wash due to its ability to generate intense physical and chemical effects through the rapid collapse of microbubbles. When spent wash passes through a cavitating device (orifice, venturi, rotor–stator), the sudden pressure fluctuations form vapor cavities that implode violently, producing localized hotspots with extremely high temperature (5000 K), pressure (>1000 atm), shear forces, and highly reactive free radicals such as $\bullet\text{OH}$. These phenomena enable HC to target and degrade the recalcitrant organic compounds present in raw spent wash particularly melanoidins, phenolics, furfurals, and long-chain polymeric organics that are resistant to biological degradation in untreated form [21].

One of the key advantages of HC is its ability to break down complex, high-molecular-weight melanoidin structures, leading to decolorization and a significant reduction in toxicity. The cleavage of chromophoric groups reduces the intense dark brown colour typical of distillery spent wash. Additionally, cavitation enhances COD solubilization, meaning a greater fraction of particulate and complex organic matter is converted to soluble, readily biodegradable forms. This directly improves the biodegradability index (BOD/COD ratio), which is crucial for efficient anaerobic digestion and methane generation [22].

HC also facilitates a noticeable reduction in toxicity by degrading phenolics, aromatic compounds, and inhibitory organics that otherwise suppress microbial activity. As a result, anaerobic microbes gain better substrate accessibility, allowing faster microbial

uptake and enhanced metabolic activity during biogas production. Improved accessibility arises from the disintegration of long-chain organics into smaller intermediates, reducing inhibition and enabling more efficient conversion into biogas [23].

Overall, hydrodynamic cavitation serves as an eco-friendly, chemical-free, and energy-efficient pretreatment technology that enhances the biodegradability of distillery spent wash, enabling higher methane yields and more stable anaerobic digestion performance.

1.11 Research Gap and Need for Present Study

Although hydrodynamic cavitation (HC) has shown considerable potential in treating high-strength industrial effluents, very limited research specifically addresses its application to molasses-based distillery raw spent wash, which contains a highly recalcitrant mix of melanoidins, phenolics, and polymeric organics. Existing studies often focus on partially treated effluents or synthetic wastewaters, leaving a gap in understanding HC performance on raw, real-field spent wash, which is far more complex and difficult to degrade. Moreover, there is a lack of systematic evaluation of critical HC process parameters—such as inlet pressure, cavitation number, number of cavitation passes or cycle times, and hydraulic retention time—despite their significant influence on pollutant degradation, solubilization, and energy efficiency. Importantly, few studies quantitatively link HC pretreatment to improvements in anaerobic digestion performance, particularly regarding methane yield, biodegradation rates, and enhancement of the BOD/COD ratio. This absence of comprehensive, parameter-optimized, and biogas-focused investigations highlights a clear research gap and underscores the need for the present study, which aims to evaluate and optimize hydrodynamic cavitation as a sustainable pretreatment strategy for enhancing biogas production from molasses-based raw spent wash.

1.12 Statement of Problem

- Anaerobic digestion of cane molasses-based distillery spent wash achieves only upto 65% COD reduction, which is the current limitation, as a large fraction of COD and refractory organics remain untreated.

- The treated effluent, known as Bio-methanated Spent Wash (BMSW) or treated spent wash (TSW), still contains residual COD, persistent refractory compounds, and intense dark brown color.
- Refractory organics in spent wash mainly include melanoidins, phenolics, and caramel-like compounds.
- Due to their long-chain and complex structure, these compounds are difficult to degrade, resist biological treatment, and contribute to the color problem. Hydrodynamic cavitation can break complex refractory molecules into smaller, more biodegradable forms and also help in partial decolourization.
- However, limited research has been done on applying hydrodynamic cavitation to study its effect on these recalcitrant organics and, it presents a promising pretreatment approach for Distillery Spent Wash.

1.13 Scope and Significance of the Study

The present study focuses on the optimization of hydrodynamic cavitation (HC) as a pretreatment strategy to enhance the biodegradability and biogas generation potential of molasses-based distillery spent wash. By systematically evaluating key HC operating parameters, the study aims to establish a scientifically robust framework of Hydrodynamic cavitation (HC) which has proven highly feasible for treating distillery spent wash, as it significantly improves the BOD/COD ratio and enhances biodegradability, directly correlating with higher methane yield during anaerobic digestion. The identification of optimal pilot-scale HC parameters supports effective scale-up, positioning HC–AD integration as a sustainable and efficient wastewater management strategy for distilleries. The adoption of HC aligns with eco-friendly and energy-efficient waste processing, contributing to broader circular economy and waste-to-energy concepts by converting a high-strength industrial pollutant into a valuable renewable energy resource. Additionally, the findings hold significant industrial relevance, as optimized HC pretreatment can support sustainable distillery wastewater management, reduce environmental pollution, minimize chemical usage, and offer a scalable, cost-effective solution for large distillery operations seeking compliance with environmental regulations and cleaner production practices.

1.14 Aim and Objectives

- To evaluate the effectiveness of hydrodynamic cavitation (HC) as a pretreatment for distillery spent wash which includes raw spent wash (RSW) and treated spent wash (TSW).
- To study the effect on HC on different key parameters like COD, TS, VS, color and methane potential after HC treatment for both RSW and TSW.
- To evaluate the improvement in substrate biodegradability based on increased biomethane potential (BMP).
- To assess the conversion of non-biodegradable COD into biodegradable fractions through enhancement in methane yield (BMP) after HC pretreatment.

Chapter 2
LITERATURE REVIEW

Literature Review

2.1 Distillery Spent Wash

Distillery spent wash is a high-strength, dark brown effluent generated from ethanol production, particularly from molasses-based distilleries. It contains extremely high COD (80,000–130,000 mg/L), BOD, color, and concentrations of recalcitrant compounds such as melanoidins, phenolics, sulfates, and other complex organics that resist biodegradation. RSW also exhibits low pH and significant total dissolved and suspended solids, contributing to its toxicity and its inhibitory effects on microbial processes. The persistent color and organic load make conventional treatment methods insufficient, necessitating pre-treatment approaches that improve solubilization and biodegradability. As a result, advanced methods like hydrodynamic cavitation are increasingly studied to enhance anaerobic digestion efficiency and reduce environmental impacts associated with distillery wastewater [24]. Various characteristics of Raw Spent Wash from various literature review are shown below in Table 2.1.

Table 2.1 Characteristics of Distillery Spent Wash from literature review

Colour	pH	Alkalinity (mg/L)	TDS (mg/L)	TSS (mg/L)	BOD (mg/L)	COD (mg/L)	References
Dark Brown	4.5-5.5	-	12,600-14,700	2,100-6,500	65,000 to 92,000	1,03,000-1,58,000	[25]
Dark Brown	2.8-3.0	-	37,500	4500	63,000	89,000	[26]
Dark Brown	4.0-4.5	3500	-	10,000	45,000-60,000	8000-1,20,000	[9]
Dark Brown	3.8-4.0	-	95,000 to 1,10,000	30,000 to 40,000	40,000 - 50,000	80,000-1,00,000	[27]
Dark Brown	3.9-4.3	-	-	-	46,100-96,000	1,04,000-1,34,000	[28]
Dark Brown	-	-	58,000-76,000	2000-14000	45,000-60,000	70,000-98,000	[29]
Dark Brown	4.0-4.5	-	-	2,400-5000	35,000-50,000	1,00,000-1,50,000	[30]
Dark Brown	3.8	-	1,12,400	16,380	-	1,62,000	[31]
Reddish Brown	3-5	-	12,000-14,000	-	30,000-40,000	75,000-80,000	[32]

2.2 Hydrodynamic Cavitation

Hydrodynamic cavitation arises from rapid pressure fluctuations within a flowing liquid, typically generated using devices such as venturi tubes, orifice plates, or rotor-stator assemblies. These fluctuations lead to the formation and collapse of microbubbles, producing localised hotspots of extremely high temperature and pressure as well as strong shear forces. Such conditions promote the disintegration of organic matter, enhanced solubilization, and the generation of reactive radicals capable of degrading recalcitrant pollutants. Initially explored for chemical intensification and industrial processing, HC has more recently emerged as a powerful pre-treatment method in wastewater management due to its energy efficiency, operational simplicity, and scalability. Studies have consistently demonstrated its effectiveness in reducing particle size, breaking complex organics, and improving biodegradability, making it a suitable approach to enhance anaerobic digestion performance [33]. Table 2.2 presents the changes in the characteristics of distillery spent wash after hydrodynamic cavitation (HC) treatment.

Table 2.2 Various changes in Characteristics of Distillery spent wash after HC

Sr. No.	Type of wastewater	Treatment type	Conditions	Findings	References
1	Distillery wastewater (biomethanated)	HC	Inlet pressure: 5 bar 5 passes	Reduced in COD by 28%, TOC by 15%, and color by 40%, while increasing soluble COD by 30%.	[5]
2	Distillery vinasse (spent wash)	Vortex-based HC	Mode of operation- Continuous recirculation system Inlet pressure range- 2–6 bar	increased soluble COD by 30%, reduced color by 40%, and decreased total COD by 20%, while improving methane yield by 25%	[1]
3	Vinasse	Vortex-based Hydrodynamic Cavitation (HC)	Inlet pressure: 3.9 bar Flow rate: 1.54 m ³ h ⁻¹ Working volume: 14 L and 1, 2, 5 passes	Additional passes showed negligible benefit Improved anaerobic digestion performance	[1]

4	Distillery spent wash	Vortex-based Hydrodynamic Cavitation (HC)	Inlet pressure: 3.9 bar Flow rate: 1.54 m ³ h ⁻¹ Working volume: 12 L and 2, 10, 20 passes (recirculation)	14% increase in BMP COD conversion to biogas increased from 59% → 65–68%	[1]
5	Industrial effluents (mixed organics)	HC / HC + AOPs	Inlet pressure: 3–6 bar Flow velocity at throat: High enough to ensure cavitation number < 1 Number of passes: 3–5 passes	HC 35% COD removal, while hybrid HC + oxidant systems 70% COD reduction and 80% removal of recalcitrant organics,	[34]
6	Waste-activated sludge	HC	Optimal pressure: 3–5 bar Passes studied: 1–5 passes	Increased sludge solubilization by 35%, reduced particle size by 50%, and improved dewaterability by 30%	[35]
7	Waste-activated sludge (rotary cavitator)	Rotary HC	Optimal RPM: 2,500 – 3,000 rpm Flow rate range: 0.5 – 2.0 m ³ /h Effective passes: 3 – 6 passes	Reduced sludge viscosity by 50–70%, decreased particle size by 40–60%, and enhanced cell disruption with a 40% increase in soluble COD,	[36]
8	Pharmaceutical wastewater (tetracycline)	HC + photocatalysis	Inlet pressure: 3–5 bar Flow regime: Cavitation number < 1 (stable cavitation) Number of passes: 3–5 passes	50% COD reduction	[37]
9	Pesticide effluent (methomyl)	HC + Ozone / HC + H ₂ O ₂	Inlet pressure: 3–6 bar Flow velocity at venturi throat: High enough to ensure stable cavitation (cavitation number < 1) Number of passes: 3–5 passes Treatment time: 20–40 minutes	65 % COD reduction	[38]

10	Refinery wastewater	HC	Inlet pressure: 1–3 bar Number of passes: 2–4 passes Equivalent treatment time: 10–30 minutes	55% COD reduction, 70% pollutant removal, 40% TOC reduction, and 70% pathogen removal,	[39]
11	Distillery spent wash	Vortex-based Hydrodynamic Cavitation (HC)	Inlet pressure: 3.9 bar Flow rate: 1.54 m ³ h ⁻¹ Working volume: 12 L and 2, 10, 20 passes (recirculation)	14% increase in BMP COD conversion to biogas increased from 59% → 65–68%	[1]
12	Cooking wastewater	HC + Fenton oxidation	Operating mode: Continuous flow with recirculation Inlet pressure: 0.3–0.6 MPa (≈ 3–6 bar) Number of passes: 4	Reduced COD by 30%, while the hybrid HC + Fenton process achieved 70% COD reduction, along with 85% phenolic removal and 45% TOC reduction	[40]

2.3 Raw Spent Wash & Treated Spent Wash

1. The literature review from the paper "Water Pollution by Sugar Mills and Distillery Spent Wash" establishes the critical environmental problem posed by Distillery Spent Wash (DSW), the highly acidic wastewater generated during alcohol production. The paper characterizes DSW as a dark, acidic effluent with extremely high Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), registering in the range of 30000-40000mg/l and 75000-80000mg/l, respectively, significantly exceeding discharge limits. The practice of discharging this untreated waste into local water bodies causes severe pollution, adversely affecting aquatic life, soil physicochemical characteristics, and crop productivity due to the high organic load. Though the spent wash is rich in plant nutrients like N, P, K, Ca, and Mg, its application requires dilution (1:10 to 1:50) to mitigate inhibitory effects and successfully leverage its value as a fertilizer. The paper concludes that proper treatment is necessary to solve the problem, with biological methods particularly anaerobic digestion being favored by the industry as a cost-effective, first-stage approach to reduce the high organic content [32].

2. Rubia, Lara Borges, S Kardos, Levente (2024) "Composting of Distillery Spent Wash" establishes Distillery Spent Wash (DSW) as a major environmental challenge due to its high acidity, significant organic load, and notable content of salts, polyphenols, and heavy metals. Given the billions of liters of DSW generated annually, its improper disposal causes severe environmental and health impacts. Composting is presented as a viable and sustainable management strategy for DSW, promoting the recycling and stabilization of organic matter and essential nutrients. The literature examines various composting methods, including single composting, co-composting, and vermicomposting, along with their respective benefits and drawbacks. The effectiveness of the process can be optimized by co-composting DSW with various materials, such as sewage sludge, animal manure, bagasse, and filter cake, which serve as sources of nitrogen and microorganisms to enhance the degradation rate and final compost quality. Key challenges in composting DSW, however, include its low pH, high salt content, imbalanced carbon-to-nitrogen ratio, and potential phytotoxicity. Ultimately, the review concludes that composting is an effective way to produce a stable, nutrient-rich organic soil amendment capable of improving crop yields and soil properties [41].
3. Sarayu Mohana, Bhavik K. Acharya, and Datta Madamwar (2009) in their review "Distillery spent wash: Treatment technologies and potential applications" highlight that Distillery Spent Wash (DSW) is a critical environmental issue, with 8-15 L of deep brown, hot, and acidic effluent generated for every liter of alcohol produced. This waste has a high BOD/COD ratio and is rich in inorganic salts. Its stubbornness to degrade stems from the presence of melanoidins, which are toxic to microorganisms and resist typical treatment. Stringent legislative mandates, like the goal for zero discharge, have intensified the need for effective technologies. The literature on DSW treatment primarily explores both biological methods (anaerobic systems like UASB and aerobic systems using fungi or bacteria) and physicochemical processes (adsorption, coagulation, and oxidation). The review suggests an ideal scheme starting with biomethanation as the primary step, followed by physicochemical and aerobic treatments. Furthermore, a significant area of research is the valorization of DSW by exploiting its nutritive potential for applications such as producing high-value compounds [3].
4. Lekshmi S.R. (2013) in the paper "Treatment and Reuse of Distillery Wastewater" highlights that the large volume of wastewater, known as Distillery Spent Wash

(DSW), generated by distillery industries in India poses a severe environmental threat due to its high COD (82,000mg/l), BOD, low pH, and dark color from recalcitrant compounds. The literature widely explores various treatment technologies and reactor configurations, but this study focuses on the performance of a Hybrid Anaerobic Baffle Reactor (HABR) for DSW treatment. The primary objective was to demonstrate the effectiveness of using anaerobic digestion as a complete solution for BOD and COD reduction, coupled with a suitable oxidation technique for post-treatment. The proposed methodology utilized a two-stage process: the first stage, anaerobic digestion, effectively reduced the initial COD from 82,000 to 17,000. Subsequently, an Advanced Oxidation Process (AOP) using hydrogen peroxide was applied to enhance the biodegradability of the remaining COD before a final anaerobic stage. This combined approach achieved an impressive overall COD removal efficiency of 92%, reducing the final COD to 6000 thus achieving the objective of producing effluent suitable for potential reuse in agriculture [42].

5. Wioleta Mikucka and Magdalena Zielińska in their review article “Distillery Stillage: Characteristics, Treatment, and Valorization” (2020), provide a comprehensive overview of the environmental problems, treatment challenges, and resource recovery opportunities associated with distillery stillage (spent wash). The authors explain that distillery stillage is an extremely high-strength wastewater containing very high COD, BOD, dark melanoidins, phenolics, nitrogen, and salts, making it one of the most difficult industrial effluents to treat. They discuss a wide range of treatment methods—including aerobic, anaerobic, physico-chemical, membrane, and thermal techniques—and highlight that no single method is fully effective, emphasising the need for integrated, multi-stage approaches. The paper also stresses the potential for valorising stillage, such as recovering polyphenols, volatile fatty acids (VFAs), and polysaccharides, aligning with modern biorefinery and circular-economy concepts. Overall, the paper concludes that more research is needed to develop efficient, economical, and sustainable technologies for both treatment and resource recovery from distillery stillage [30].

2.4 Hydrodynamic Cavitation

6. Jitendra Carpenter, Mandar Badve, Sunil Rajoriya, Suja George, Virendra Kumar Saharan, and Aniruddha B. Pandit (2016) in their review "Hydrodynamic cavitation:

an emerging technology for the intensification of various chemical and physical processes in a chemical process industry" establish that Hydrodynamic Cavitation (HC) has been extensively explored since the 1990s as a novel, energy-efficient technique for process intensification. HC is defined as the phenomenon where pressure variations in a flowing fluid cause the formation, growth, and subsequent violent collapse of microbubbles or cavities. This collapse generates "hot spots" with extremely high local temperatures (1,000-10,000~K) and pressures (100-1,000bar), which drive chemical and physical transformations. The review details that HC can be generated by various constricting devices, such as the most widely used orifice plates and Venturis, or by high-speed homogenizers. Unlike acoustic cavitation (AC), HC generates more cavities due to its geometrical configuration, and it is found to produce both stable and transient cavitation, which can be controlled based on the device geometry. The paper comprehensively discusses HC applications, including wastewater treatment, microbial cell disruption, biodiesel synthesis, and the production of nanomaterials [43].

7. Anteneh Mesfin Yeneneh, Khadija Al Balushi, Tahereh Jafary, and Amjad Said Al Marshudi in their review "Hydrodynamic Cavitation and Advanced Oxidation for Enhanced Degradation of Persistent Organic Pollutants: A Review" assert that conventional wastewater treatment technologies struggle to meet sustainability requirements when dealing with municipal and industrial effluents, particularly due to the presence of Persistent Organic Pollutants (POPs). They position Hydrodynamic Cavitation (HC) as an emerging, energy-efficient, and sustainable technique for process intensification². The HC phenomenon involves the formation and violent collapse of vapor-filled cavities, which generates intense physical effects (shockwaves, turbulence) and chemical effects through the production of highly reactive hydroxyl radicals. The literature reviewed proves that HC is effective for the enhanced degradation of persistent pollutants, but its efficacy is significantly improved through synergistic coupling with Advanced Oxidation Processes (AOPs). This combination helps overcome the limitations of biological processes, which are often hindered by recalcitrant compounds. The review provides an overview of research on operational parameters, model-based studies, and a feasibility analysis to recommend options for the pilot-to-large-scale implementation of HC and AOPs to ensure better sustainability in wastewater treatment [2].

8. Niveditha Asaithambi, Poonam Singha, Madhuresh Dwivedi, and Sushil K. Singh (2019) in their review "Hydrodynamic cavitation and its application in food and beverage industry: A review" establish Hydrodynamic Cavitation (HC) as a critical non-thermal technology for process intensification in the food sector. HC is defined by the energy release from the implosion of vapor bubbles, which occurs due to localized pressure changes in a flowing liquid, and is being harnessed to perform various physical and chemical tasks. The literature identifies the primary applications of HC in the food and beverage industries as a tool for homogenization, pasteurization, breaking, or mixing food macromolecules. Unlike thermal processing, HC can minimize degradation of heat-sensitive components, preserving the nutritional and functional properties of food. The review discusses the fundamental principles of HC, including the design aspects of different cavitators like orifice plates and Venturis, and details its uses in cell disruption, emulsification, extraction of bioactive compounds, and microbial inactivation (non-thermal pasteurization). The authors also perform comparative studies, emphasizing the advantages of HC over acoustic cavitation, particularly in terms of energy efficiency and scalability for industrial-level food processing [44].
9. Haoxuan Zheng, Ying Zheng, and Jesse Zhu (2022) in their review "Recent Developments in Hydrodynamic Cavitation Reactors: Cavitation Mechanism, Reactor Design, and Applications" establish Hydrodynamic Cavitation (HC) as a promising technology for process intensification due to its high energy efficiency, cost-effective operation, and scale-up possibilities. The fundamental phenomenon is characterized by the formation, growth, and violent collapse of bubbles, which releases a large amount of local energy. The paper provides an extensive review, covering the fundamentals of HC and its main variables that support applications in radical-induced chemical reaction processes. The authors categorize and compare two primary reactor types: stationary and rotational HC reactors. They discuss the optimization of design parameters and reactor performance at both laboratory and pilot scales. The review confirms that the unique features of HC, including the generation of hydroxyl radicals, are widely applied to various fields. Key applications include oxidation reactions, wastewater treatment, and physical processes such as emulsion generation and component extraction [45].
10. Cuiwei Liu, Xuejie Li, Anqi Li, Zhaoxue Cui, Lei Chen, and Yuxing Li (2020) in their paper "Cavitation onset caused by a dynamic pressure wave in liquid pipelines",

address a critical shortcoming in the conventional prediction of cavitation in liquid pipelines. The literature conventionally uses a velocity-based Cavitation Number (CN), C , where cavitation is expected when $C \leq 1$. However, prior experimental observations and incidents of pipe bursts suggest that this conventional CN often incorrectly predicts the onset of cavitation. This misprediction is particularly evident in scenarios where a dynamic pressure wave with a small amplitude—caused by events such as valve opening or pipeline leakage induces bubble formation even when the velocity-based criterion is not met. To solve this, the authors propose a new Cavitation Number that is specifically based on the amplitude of the dynamic pressure wave. This dynamic-pressure-based CN is intended to provide a more reliable methodology for ascertaining the onset of cavitation in liquid transport systems subjected to pressure fluctuations. The paper's main contribution is centered on verifying the validity of this new dynamic CN through experimental evidence [46].

11. Giuseppe Mancuso, Michela Langone, and Gianni Andreottola (2020), in their review "A critical review of the current technologies in wastewater treatment plants by using hydrodynamic cavitation process: principles and applications", assert that Hydrodynamic Cavitation (HC) has become increasingly significant in wastewater treatment over the last decade. The core of HC involves the formation, growth, and violent collapse of vapour cavities, which release powerful energy due to rapid local pressure variations. This energy provides both mechanical effects (used for cell disintegration of microorganisms) and, crucially, oxidative capability (used to treat organic, toxic, and bio-refractory contaminants). The literature extensively defines the fundamental principles and analyses the governing mechanisms of both HC generation and pollutant degradation. The review critically assesses how various operational and fluid parameters influence the effectiveness of HC and details methods for HC process intensification. Finally, the paper concludes by summarizing the key applications of HC, either alone or in synergistic combination with other processes, within the wastewater treatment sector [47].
12. Kashyap P. Mishra and Parag R. Gogate (2010) in their paper "Intensification of degradation of Rhodamine B using hydrodynamic cavitation in the presence of additives" address the need for advanced methods to treat waste dye effluents, which are often highly coloured and refractory. The literature confirms the application of Hydrodynamic Cavitation (HC) as an effective technique, capitalizing on the high energy released during bubble collapse to generate reactive species like hydroxyl

radicals for pollutant degradation. The core of this study involves comparative testing of two key HC reactor configurations, the orifice plate and the Venturi tube, for the degradation of Rhodamine B dye. Furthermore, the work extensively reviews the use of HC in combination with various intensifying additives, such as Hydrogen Peroxide and Fenton's reagent, to significantly boost the rate of degradation. The literature also investigates the critical impact of various operating parameters, operating temperature, and inlet pressure, on the overall efficiency of the dye removal process [48].

13. Prof. Yogesh. S. Khandekar and Dr Narendra. P. Shinkar, in their review "Treatment Methods of Distillery Spent Wash: A Review", establishes that grain-based distilleries are classified as a "Red Category" industry primarily due to the immense volume (12-15 litres per litre of alcohol) and high strength of the wastewater generated. The spent wash exhibits extremely high pollutant loads, with Biological Oxygen Demand (BOD) ranging from 40,000-65,000 mg/lit. and Chemical Oxygen Demand (COD) from 80,000-1,40,000 mg/lit. This high organic and inorganic impurity content, including chlorides, sulphates, phosphates, potassium, and calcium, poses a critical environmental threat, capable of causing the eutrophication of natural water bodies. The recalcitrant and dark-colored nature of the effluent is specifically attributed to the presence of complex compounds like melanoidins, ceramal, polyphenols, and various sugar decomposition products. Consequently, the paper reviews a variety of essential techniques—including biological, thermal, and incineration methods—employed to treat this complex distillery effluent [49].
14. Zhilin Wu, Francisco J. Yuste-Córdoba, Pedro Cintas, Zhansheng Wu, Luisa Boffa, Stefano Mantegna, and Giancarlo Cravotto (2018) in their paper "Effects of ultrasonic and hydrodynamic cavitation on the treatment of cork wastewater by flocculation and Fenton processes" establish that Cork Wastewater (CW) poses a significant environmental challenge due to its high content of toxic and refractory organic pollutants, specifically phenolic compounds. The literature highlights the necessity of removing these inhibitory substances before conventional biological treatment is feasible. The study rigorously compares the performance of two advanced oxidation techniques: Ultrasonic Cavitation (US) and Hydrodynamic Cavitation (HC). It is found that both US and HC are effective strategies for the environmental remediation of CW. Crucially, the paper demonstrates that using cavitation (both US and HC) as a pre-treatment for Fenton oxidation (FE) significantly enhances the degradation efficiency, leading to a substantial reduction in both Chemical Oxygen Demand (COD) and

polyphenol (PP) content. The key finding is the clear synergistic benefit of combining a physical process (cavitation) with a chemical process (Fenton oxidation) to overcome the recalcitrant nature of the pollutants in CW [50].

15. Janez Kosel and co-authors in their paper “Cavitation as a Potential Technology for Wastewater Management – An Example of Enhanced Nutrient Release from Secondary Pulp and Paper Mill Sludge” (2019) investigate how hydrodynamic cavitation can be used as an efficient pretreatment method for improving sludge disintegration and nutrient release in the pulp and paper industry. The study tests two cavitation systems— a Venturi-based blow-through reactor and a rotating cavitation generator at both lab-scale and pilot-scale. Results show that cavitation significantly increases soluble COD, nitrogen, and phosphorus due to microbial cell rupture and breakdown of organic flocs, with the rotating device demonstrating higher efficiency. The authors also highlight that combining cavitation with alkaline pretreatment (NaOH) further enhances nutrient release and improves economic feasibility. Overall, the paper concludes that hydrodynamic cavitation is a cost-effective, scalable, and energy-efficient technology that can significantly improve wastewater management in industrial sludge treatment [51].
16. P. Mallick, J.C. Akunna, and G.M. Walker, in their paper “Anaerobic digestion of distillery spent wash: Influence of enzymatic pre-treatment of intact yeast cells” (Bioresource Technology, 2010) investigate how enzymatic hydrolysis of intact yeast cells improves the anaerobic digestion efficiency of distillery residues such as spent wash and pot ale. The study shows that yeast cell walls, composed of glucans and mannoproteins, limit biodegradation, making hydrolysis a crucial step. Enzymes like β -glucanase combined with protease achieved up to 90% yeast cell lysis, significantly enhancing organic breakdown. When enzymatically pretreated, spent wash and pot ale showed major improvements in COD reduction—up to 87%, compared to only 13% without pretreatment. The results clearly demonstrate that intact yeast cells are a rate-limiting factor in anaerobic digestion and that enzymatic pretreatment greatly enhances bioconversion efficiency and methane potential [52].
17. Anupam Mukherjee, Aditi Mullick, Ravi Teja, Pavani Vadthya, Anirban Roy, and Siddhartha Moulik in their paper “Performance and energetic analysis of hydrodynamic cavitation and potential integration with existing advanced oxidation processes: A case study for real-life greywater treatment” (Ultrasonics Sonochemistry, 2020) evaluate the efficiency of hydrodynamic cavitation (HC) and its integration with

advanced oxidation processes (AOPs) for treating kitchen greywater. The study optimizes cavitation device geometry, inlet pressure, pH, and dilution ratio, showing that HC alone gives moderate COD/TOC reduction but significantly improves when combined with oxidants. The authors report that integrating HC with H₂O₂ and O₃ leads to the highest COD (98.25%) and TOC (76.26%) removal, with improved reaction rates, energy efficiency, and reduced toxicity (as shown by algal viability tests). Overall, the paper concludes that hybrid HC–AOP systems are highly efficient, economical, and scalable for real wastewater treatment [53].

18. Deepak Pant and Alok Adholeya in their review paper “Biological approaches for treatment of distillery wastewater: A review” (Bioresource Technology, 2007) provide a comprehensive overview of the pollution characteristics of molasses-based spent wash and summarize the performance of various biological treatment methods. The authors highlight that distillery spent wash is extremely difficult to treat because of its high COD/BOD, acidic pH, dark melanoidin pigments, phenolics, sulphur compounds, and toxicity to aquatic life. The review details anaerobic systems such as UASB and hybrid reactors, which can remove up to 90% COD, but still fail to eliminate colour and phenolics. They also discuss aerobic bacterial, fungal and mixed-consortium treatments, showing that white-rot fungi and specific bacterial strains can decolourize melanoidins but often need nutrient supplements and diluted effluent. The authors conclude that no single biological process is sufficient, and effective treatment requires a combination of anaerobic + aerobic + specialized decolourization steps [4].
19. K.V. Padoley, Virendra Kumar Saharan, S.N. Mudliar, R.A. Pandey, and Aniruddha B. Pandit in their paper “Cavitationally Induced Biodegradability Enhancement of a Distillery Wastewater” (Journal of Hazardous Materials, 2012) evaluate hydrodynamic cavitation (HC) as an effective pretreatment for biomethanated distillery wastewater. The study shows that HC significantly improves biodegradability by increasing the BOD₅/COD ratio from 0.14 to 0.32, mainly due to breakdown of complex melanoidin-rich organics. The authors also report notable reductions in COD (32%), TOC (31%), and colour (up to 48%) under optimised conditions. Importantly, biochemical methane potential tests show a 6-fold increase in biogas generation (400 mL vs. 60 mL in control), demonstrating HC’s strong ability to enhance anaerobic digestion. Overall, the paper concludes that hydrodynamic cavitation is a promising, scalable pretreatment technology for reducing toxicity and improving biodegradability of recalcitrant distillery effluent [5].

20. Sanjay Nagarajan and Vivek Vinayak Ranade in their paper “Pre-treatment of distillery spent wash (vinasse) with vortex-based cavitation and its influence on biogas generation” (2020) investigate, for the first time, the use of a vortex-based hydrodynamic cavitation (HC) device to pretreat spent wash and vinasse for enhanced biogas production. The authors show that passing high-COD spent wash through the HC device (2 passes at 3.9 bar) results in a 14% increase in biomethane potential (BMP), along with faster digestion (shorter lag phase) and breakdown of recalcitrant organics. According to the results in Figure 2(a), HC pretreatment eliminates diauxic digestion and makes the COD more accessible for microbes. Industrial trials shown in Figure 3 demonstrate 10–22% higher biogas yields in large-scale digesters after just one HC pass, confirming scalability. The study concludes that vortex-based HC offers a net energy gain (1 GJ/ton COD) and is a promising, energy-efficient pretreatment for distillery wastewater [1].
21. Pooja Thanekar and Parag Gogate in their review paper “Application of Hydrodynamic Cavitation Reactors for Treatment of Wastewater Containing Organic Pollutants: Intensification Using Hybrid Approaches” (2018), provide a comprehensive overview of how hydrodynamic cavitation (HC) and its hybrid combinations can effectively degrade various organic pollutants. The paper explains how HC generates extreme localized conditions—high temperatures, intense shear forces, and hydroxyl radicals that break down dyes, pesticides, pharmaceuticals, phenolics, and other refractory organics. They highlight that although HC alone often achieves partial degradation, combining it with oxidants such as H_2O_2 , ozone, UV radiation, Fenton reagent, or photocatalysis results in significantly enhanced treatment efficiency due to synergistic radical formation. The authors also outline optimized operating conditions, including inlet pressure, pH, fluid velocity, and temperature, all of which strongly influence the extent of pollutant degradation. They report that hybrid HC systems achieve higher COD and TOC reductions, faster mineralization rates, and improved reaction kinetics, making them highly suitable for industrial wastewater applications. Additionally, the review emphasizes that HC reactors, especially Venturi, orifice, and rotating cavitation devices, offer scalable and energy-efficient operation compared to ultrasonic cavitation. The paper further discusses how reactor geometry and cavitation number determine cavitation intensity, directly affecting pollutant breakdown. Importantly, they highlight that HC-based hybrid processes often reduce chemical consumption and operational costs, making them more sustainable. Overall,

the review positions hydrodynamic cavitation as a promising, eco-friendly, and industrially viable intensification technology for advanced wastewater treatment and pollutant degradation.

22. A.R. Warade, along with R.W. Gaikwad, R.S. Sapkal, and V.S. Sapkal, in their paper “Review on Wastewater Treatment by Hydrodynamic Cavitation” (2016), present a detailed overview of how hydrodynamic cavitation (HC) can be effectively used for wastewater treatment. The authors explain that HC produces extremely high temperatures and pressures during bubble collapse, creating strong oxidizing radicals such as $\bullet\text{OH}$ that rapidly break down complex organic pollutants. The review highlights significant advantages of HC over ultrasonic cavitation and traditional advanced oxidation processes, particularly its lower operational cost, easier scale-up, and higher cavitation efficiency. They also discuss how parameters like inlet pressure, cavitation number, orifice/venturi geometry, and turbulence intensity play a key role in determining pollutant degradation efficiency. Warade and co-authors further emphasize that HC has shown successful removal of dyes, phenols, pesticides, and other recalcitrant contaminants, making it suitable for a wide range of industrial effluents. The review also outlines the strong potential of hybrid HC systems such as HC combined with H_2O_2 , ozone, Fenton reagent, or UV because these combinations provide synergistic radical generation and enhanced reaction rates. Another important contribution of the paper is its discussion on energy efficiency, noting that HC requires relatively low energy input compared to other oxidation technologies. The authors also highlight that HC avoids chemical sludge generation, thereby reducing secondary disposal issues. They conclude that hydrodynamic cavitation is a promising, scalable, eco-friendly, and industrially viable technology for modern wastewater treatment challenges [54].
23. Mandar Badve, Parag Gogate, Aniruddha Pandit, and Levente Csoka in their paper “Hydrodynamic cavitation as a novel approach for wastewater treatment in wood finishing industry” (2013) examine the effectiveness of a stator–rotor hydrodynamic cavitation (HC) reactor for treating real wood-finishing wastewater. Their results show that HC alone achieves notable pollutant degradation, with COD removal reaching 49% at the optimal rotor speed of 2200 RPM. The authors also demonstrate that introducing hydrogen peroxide (H_2O_2) significantly boosts oxidation efficiency, leading to approximately 80% COD reduction due to intensified OH radical formation. Residence time is shown to be critical, with an optimal number of cavitation passes

yielding 56% COD removal. In addition to pollutant reduction, the study highlights that HC improves color removal, solubilization of organic compounds, and overall effluent biodegradability, making downstream biological treatment more effective. The authors emphasize that the stator–rotor design provides higher cavitation intensity compared to conventional devices, improving mass transfer and radical generation. They also note that HC offers operational advantages such as lower energy consumption, no chemical sludge formation, and simple mechanical scalability for industrial use. Overall, the study establishes hydrodynamic cavitation—especially in hybrid mode with H_2O_2 as a promising, eco-friendly, and highly efficient technology for treating complex industrial wastewater streams [55].

24. Yongxing Song, Ruijie Hou, Weibin Zhang and Jingting Liu in their review paper “Hydrodynamic cavitation as an efficient water treatment method for various sewage – A review” (2022) present a detailed assessment of hydrodynamic cavitation (HC) as a promising technology for diverse wastewater treatment applications. They describe how HC produces extremely high temperatures (up to 4600 K), intense shockwaves, and strong microjets during bubble collapse, leading to the formation of highly reactive hydroxyl radicals ($\bullet\text{OH}$) that drive organic degradation and microbial destruction. The authors summarize HC performance across domestic sewage, dye effluents, phenolic wastewater, and industrial streams, demonstrating consistent improvements in COD removal, colour reduction, pathogen inactivation, and mineralization. They also emphasize the superior efficiency of hybrid approaches such as $\text{HC} + \text{H}_2\text{O}_2$, $\text{HC} + \text{O}_3$, $\text{HC} + \text{Fenton}$, and $\text{HC} + \text{UV}$, which significantly enhance oxidation rates and treatment outcomes. The paper further compares commonly used cavitation reactors Venturi, orifice plate, and rotor–stator devices highlighting that rotor-based reactors generate higher cavitation intensity and deliver better pollutant degradation at lower energy input. The authors also note practical advantages of HC, including low chemical demand, easy scale-up, and high energy efficiency, making it suitable for decentralized and industrial wastewater systems. Additionally, they identify key research needs such as optimizing reactor geometry, improving mechanistic modelling, reducing energy consumption, and integrating HC with advanced materials like catalysts and photocatalysts. Overall, the review establishes HC as a powerful, scalable, and eco-friendly technology for modern wastewater treatment challenges.
25. Kenneth S. Suslick and David J. Flannigan, in their influential review “Inside a Collapsing Bubble: Sonoluminescence and the Conditions During Cavitation” (2008),

explore the extreme physical and chemical phenomena created during bubble collapse in acoustic cavitation. They explain that cavitation focuses mechanical energy into microscopic hotspots, generating extraordinarily high temperatures (up to 20,000 K) and pressures of several thousand atmospheres. These conditions produce sonoluminescence, a light-emission event that serves as a diagnostic signature of bubble implosion. The authors describe how these extreme thermodynamic conditions lead to the formation of reactive species, particularly hydroxyl radicals, which drive powerful oxidation reactions in liquids. Through spectroscopic analysis, they distinguish between single-bubble and multibubble cavitation, noting that the latter is more chemically relevant for practical applications. They also discuss how shockwaves, microjets, and rapid compression contribute to chemical bond breakage and molecular fragmentation. Importantly, Suslick and Flannigan highlight the broad relevance of cavitation to sonochemistry, advanced oxidation processes, nanoparticle synthesis, and environmental remediation. Their review establishes cavitation as a unique and highly efficient method for inducing extreme localized conditions, laying the scientific foundation for modern cavitation-based wastewater treatment technologies [56].

2.5 Biochemical Methane Potential (BMP)

26. The study by Jeebon et al. demonstrates that distillery spent wash (DSW) can be efficiently converted into biomethane through anaerobic digestion when key parameters such as inoculum-to-substrate ratio and substrate composition are optimized. An optimal I/S ratio of 2:1 significantly improved gas production, and codigestion of substrates enhanced microbial activity and nutrient balance; Jeebon et al. emphasize that such combinations lead to better performance than single substrates. The highest biogas yield (310.97 NmL) was achieved using a combination of cow dung and baker's yeast, indicating a strong synergistic effect, while baker's yeast alone also showed high efficiency. The study further highlights that a mixture of DSW, cow dung, and kitchen waste provides an ideal C/N ratio (20–30) for stable digestion, whereas materials like dry leaves and rice straw are less effective due to their high C/N ratios; as noted by Jeebon et al., these substrates contribute minimally to overall methane yield. Additionally, biogas production increased steadily over time, with the best results observed in substrate mixtures rich in spent wash and kitchen waste, and the

type of inoculum significantly influenced performance. Overall, Jeebon et al. conclude that optimized anaerobic digestion of DSW is a promising and sustainable solution for both waste management and renewable energy generation [57].

27. Lalitha et al. demonstrate that high TDS distillery spent wash can be effectively treated through biomethanation, converting it into a valuable source of renewable energy while reducing environmental pollution. The study highlights that spent wash, rich in organic matter, can generate methane via anaerobic digestion involving stages such as hydrolysis, acidogenesis, acetogenesis, and methanogenesis. Maintaining an optimal pH range of 5.4–6.5 was found to enhance microbial activity and improve the production of acetate, organic acids, and biogas components like methane and CO₂. The findings also indicate that biomethanation significantly reduces COD, BOD, and acidity, making the effluent safer for disposal. Additionally, the treated residue can be used as compost or biofertilizer, supporting sustainable waste utilization. Overall, the study concludes that this process is an efficient, eco-friendly, and economically viable solution for distillery wastewater management and energy recovery [58].

Overall, the reviewed literature clearly demonstrates that hydrodynamic cavitation, either as a standalone pre-treatment or in combination with advanced oxidation processes, consistently enhances solubilization, reduces COD and color, and significantly improves biodegradability and methane yield of distillery spent wash, thereby establishing HC as an energy-efficient, scalable, and promising intensification strategy for sustainable distillery wastewater management.

Chapter 3

METHODOLOGY

3. METHODOLOGY

3.1 Materials

The Raw Spent Wash (RSW) and Treated Spent Wash (TSW) were collected from a distillery near Solapur of C heavy grade.

3.2 Introduction

This chapter focuses on various tests and methodologies that are done on Raw Spent Wash and Treated Spent Wash.

3.1.2 Treated Spent Wash (TSW)

Treated spent wash is the wastewater obtained after subjecting raw distillery spent wash to suitable physical, chemical, and biological treatment processes to reduce its extremely high pollution load. Raw spent wash is characterized by very high COD, BOD, dark colour, strong odour, acidity, and toxic organic compounds, whereas treatment processes such as pre-treatment (including dilution, pH correction, hydrodynamic cavitation, or oxidation) followed by anaerobic and aerobic biological treatment significantly reduce organic matter and toxicity. As a result, treated spent wash shows much lower COD and BOD values, improved biodegradability, reduced odour and colour intensity, and enhanced environmental safety. Depending on the level of treatment and regulations, it can be subjected to composting or to Zero Liquid Discharge.



Figure 3.1.2 Treated Spent Wash

3.1.3 Raw Spent Wash (RSW)

The sugarcane industry produces a liquid waste known as Raw Spent wash. Raw Spent wash contains a lot of inorganic and organic materials. Additionally, RSW has adequate levels of micronutrients (zinc, copper, iron, and manganese) and macronutrients

(nitrogen, phosphorus, potassium, nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulphur (S), all of which enhance crop growth and yield. A by-product of the sugar extraction process, molasses is the most widely utilized raw material in distilleries for the production of bioethanol. Following alcohol distillation, a significant amount of highly coloured spent wash is produced, which has a high organic load (80,000–150,000 mg/l for chemical oxygen demand and 40,000–60,000 mg/l for biochemical oxygen demand), is dark brown in colour, has a low pH, and is heated to a high temperature.



Figure 3.1.3 Raw Spent Wash

3.1.4 Equipment Used

The study involves usage of various equipment which are listed below:

3.1.4.1 Biochemical Methane Potential (BMP) Instrument

A BMP device enables the determination of a biomass substrate's dynamic degradation profile as well as its actual biogas and methane potential. This enables the effective screening for suitable pre-treatment techniques, the recognition of ideal retention periods and substrate combinations for co-digestion, the evaluation of the biological performance of specific digesters or process configurations, and the determination of the necessity for additives.

The Automatic Methane Potential Test System (AMPTS) BMP instrument is a specialized system designed to measure the Biochemical Methane Potential (BMP) of organic substrates subjected to anaerobic digestion. It provides accurate and completely automated measures of methane generation by tracking gas output using various feedstocks. The BMP system maintains ideal temperature, pressure, and mixing conditions throughout digestion to guarantee accurate and high-quality data. It is

designed for both industrial and research applications. It is frequently used to measure methane yields from various organic materials in waste management, renewable energy research, and biogas production studies. This assists with substrate selection and biogas plant performance optimization.

In the first stage of the process, inoculum and substrate are incubated at anaerobic conditions. The thermostatic water bath allows to incubate different temperatures. Biogas is produced (i.e. CO₂ and CH₄). The second stage is the Absorption Unit, where Biogas goes to the CO₂ absorption unit, where CO₂ is absorbed by an alkaline solution (NaOH). CH₄ remains and goes to our detection unit. The third stage is the Detection Unit, where Gas (CH₄) enters the flow cell unit and is measured. The data is stored in the BPC AMPTS software and can be accessed via a computer or internet connection.



Figure 3.1.4.1 AMPTS Instrument

3.1.4.2 pH Meter

A pH meter is used to accurately measure the acidity or alkalinity (pH) of a solution. The pH of the samples was measured using a calibrated pH meter. The instrument was first switched on and calibrated with standard buffer solutions of pH 4.0, 7.0, and 10.0. The electrode was rinsed with distilled water and gently blotted dry before use. It was then immersed in the Raw Spent Wash (RSW) and Treated Spent Wash (TSW) samples, ensuring that the sensing bulb was completely submerged. The solution was stirred gently and the reading was allowed to stabilize before recording the pH value displayed on the meter. After each measurement, the electrode was rinsed with distilled water and stored in the recommended storage solution.



Figure 3.1.4.2 pH and Conductivity Meter

3.1.4.3 Conductivity Meter

Conductivity is used in hydrodynamic cavitation to measure the dissolved ionic content in wastewater. During cavitation, complex organic compounds break into smaller molecules and ionic species, which can increase conductivity. It provides a quick and instant indication of oxidation and treatment progress. Hence, EC helps in monitoring the efficiency of the HC process. The electrical conductivity (EC) was measured using a conductivity meter calibrated with a standard 0.01 KCl solution (1.132 mS/cm). The probe was rinsed with distilled water and immersed in the RSW and TSW samples. After the reading stabilized, the conductivity value ($\mu\text{S}/\text{cm}$) was recorded.

3.1.4.4 ORP Meter

ORP (Oxidation Reduction Potential) is used in hydrodynamic cavitation to measure the oxidizing strength of the system. During cavitation, collapsing bubbles generate hydroxyl radicals ($\bullet\text{OH}$) and other oxidizing species, which increase ORP. A higher ORP indicates stronger oxidation and better pollutant degradation. Thus, ORP helps monitor the effectiveness of the cavitation treatment process. The oxidation–reduction potential (ORP) of the samples was measured using an ORP meter. The instrument was switched on and the calibration was checked using a standard ORP solution. The electrode was rinsed with distilled water and gently blotted dry before measurement. It was then immersed in the sample, ensuring that the sensing tip was fully submerged. The sample was stirred gently and the reading was allowed to stabilize, after which the ORP value (in mV) displayed on the meter was recorded.



Figure 3.1.4.4 ORP Meter

3.2.4 Characterization

3.2.4.1 Total Solids (TS)

Using a pre-weighed dish, a well-mixed sample is evaporated and dried to a constant weight in an oven set at 103–105°C. The residue that remains after drying represents the substrate's overall solids content. The weight of the residue in relation to the wet sample is then used to compute the total solids.

$$\% \text{ Total Solids} = \frac{A - B}{C - B} * 100$$

where,

A = weight of dried residue + dish (g)

B = weight of dish (g)

C = weight of wet sample + dish (g)

D = weight of residue + dish after ignition (g)



Figure 3.2.4.1 Hot Air Oven

3.2.4.2 Volatile Solids (VS)

The residue obtained from the total solid's determination is ignited at 550°C until a constant weight is achieved. The volatile solids are represented by the weight loss during ignition, and the fixed, dissolved, or suspended inorganic components are represented by the residual ash. The standard formula is then used to compute total volatile solids. This value is crucial because it provides an accurate indication of the amount of organic matter in the waste, which is necessary for evaluating and controlling biological treatment procedures.

$$\% \text{ Volatile Solids} = \frac{A - D}{A - B} * 100$$



Figure 3.2.4.2 Muffle Furnace

3.2.4.3 Total Dissolved Solids (TDS)

Total Dissolved Solids (TDS) represent the fraction of solids that pass through a standard glass fibre filter and remain in the filtrate. Organic matter, dissolved inorganic salts, and other soluble materials found in the substrate make up the majority of TDS. To fully understand the material's dissolved load, which affects anaerobic digestion efficiency and process stability, TDS measurement is crucial.

For TDS analysis, a well-mixed sample is first filtered through a standard glass fibre filter to remove suspended solids. Following that, a predetermined volume of the filtrate is put into an evaporating dish and dried at 103–105°C in a hot air oven until a constant weight is achieved. The total amount of dissolved solids in the sample is indicated by the rise in dish weight during drying.

$$\% \text{ Total Dissolved Solids (TDS)} = \frac{A-B}{V} \times 1000$$

Where,

A = Weight of dried residue + dish (mg)

B = Weight of empty dish (mg)

V = Volume of sample filtered (mL)

3.2.4.4 Total Suspended Solids (TSS)

Total Suspended Solids (TSS) represent the fraction of solids that are retained on a standard glass fibre filter and are not dissolved in the liquid phase. Organic particles and inorganic matter make up the majority of suspended solids, which have an important influence on the substrate's biodegradability and digestion kinetics during anaerobic treatment.

For TSS determination, a known volume of a well-mixed sample is filtered through a pre-weighed glass fibre filter. Following that, the filter with the residual solids is dried at 103–105°C in an oven until it reaches a consistent weight. The amount of suspended particles in the sample is indicated by the increase in filter weight during drying.

$$\% \text{ Total Suspended Solids (TSS)} = \frac{A-B}{V} \times 1000$$

Where,

A = Weight of dried filter + Residue (mg)

B = Weight of clean filter paper (mg)

V = Volume of sample filtered (mg)



Figure 3.2.4.4 Centrifuge Used for TDS and TSS

3.3 Analysis

3.3.1 Chemical Oxygen Demand (COD)

The Chemical Oxygen Demand (COD) test is commonly performed using the closed reflux dichromate method to determine the amount of oxidizable organic matter in a water or wastewater sample. A measured volume of the well-mixed sample is taken in a COD digestion vial containing potassium dichromate, sulfuric acid, and catalysts such as silver sulfate, with mercuric sulfate added to eliminate chloride interference. The vial is tightly capped, mixed, and digested in a COD reactor at about 150 °C for two hours. During digestion, organic matter is chemically oxidized by dichromate. After cooling to room temperature, the COD value is determined either by titration with ferrous ammonium sulfate or directly using a COD spectrophotometer. The result is expressed as mg/L of oxygen, indicating the strength of organic pollution in the sample.

Chemical Oxygen Demand (COD) is a measure of the amount of oxygen required to chemically oxidize the organic and inorganic matter present in the substrate. Since COD gives an estimate of the overall biodegradable and non-biodegradable organic load, it is an essential indicator for assessing the material's organic strength and suitability for anaerobic digestion.

The COD of the sample was calculated using the following equation,

$$\text{COD (mg/L)} = \frac{B - B.R \times 0.1N \times 8000}{\text{Vol of Sample}}$$

Where,

B = blank

B.R = Burette Reading



Figure 3.3.1 COD Digester

3.4 Hydrodynamic Cavitation Plant

A pilot-scale hydrodynamic cavitation setup was established at CERIPL, Butibori Laboratory. Figure 3.4 shows the HC setup.



Figure 3.4 (a) Hydrodynamic Cavitation Pilot Plant

The hydrodynamic cavitation setup for the treatment of raw spent wash and treated spent wash was operated by first storing the effluent in a 200-L feed tank, which supplied a continuous and well-mixed flow to the centrifugal pump connected to a venturi cavitation device designed to handle RSW and TSW. The pump forces the liquid through the venturi, creating the necessary pressure drop to induce cavitation, while the recirculation loop allows the effluent to pass through the cavitation zone multiple times, thereby controlling treatment duration. The system typically operates at an inlet pressure

of 1–2 bar and a flow rate of 50-75 L/min, with the number of passes adjusted between 1 to 4. This ensures sufficient cavitation activity for effective breakdown of complex organics and enhancement of wastewater biodegradability.

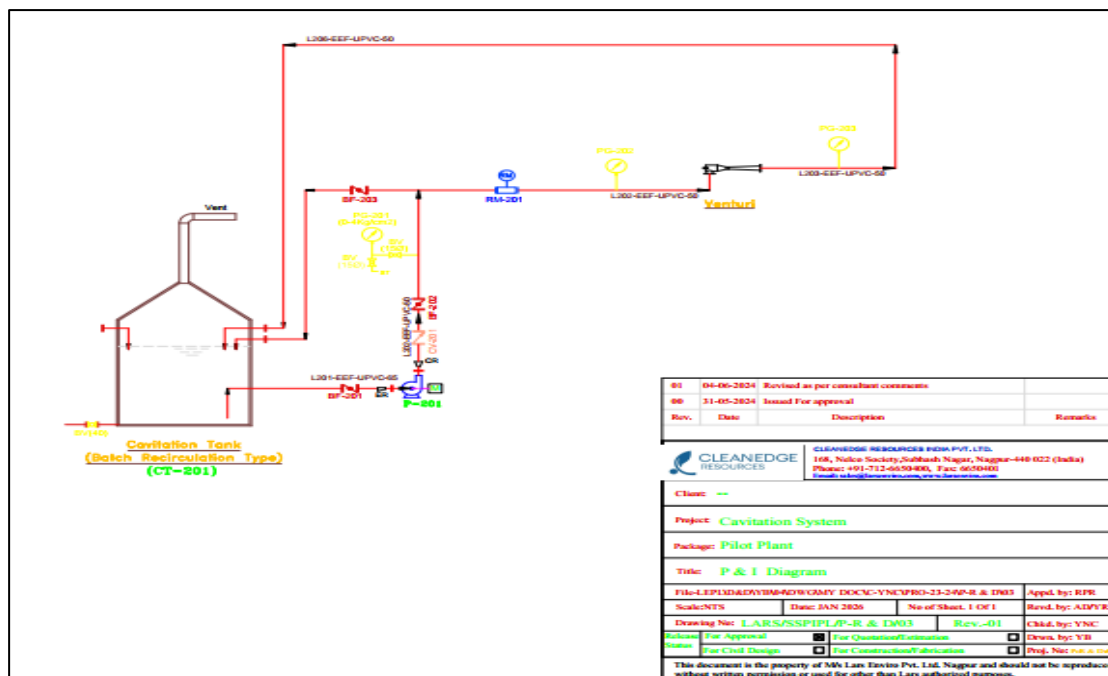


Figure 3.4 (b) P&I Diagram

3.4.1.1 Centrifugal Pump

The centrifugal pump is used for supplying the required pressure and flow rate needed to initiate cavitation. It accelerates the liquid using a rotating impeller, increasing its kinetic energy and pushing it through the cavitation device.

3.4.1.2. Cavitation Device (Venturi)

The cavitation device is the heart of the HC system. As liquid flows through a narrow throat or constriction, its velocity increases sharply, causing a sudden drop in pressure below the liquid's vapour pressure. This pressure reduction leads to the formation of vapour cavities (bubbles). When the liquid moves past the throat into a wider section, the pressure rises again, and the bubbles collapse violently. This bubble collapse generates intense micro-level forces, including shock waves and free radicals, which drive pollutant degradation.

3.4.1.3. Recirculation Loop

The recirculation loop allows the RSW and TSW to pass repeatedly through the cavitation device so that the treatment intensity can be controlled. After each pass, the liquid returns to a feed tank and is pumped again through the system. Increasing the number of passes increases the exposure of the effluent to cavitation forces.

3.4.1.4. Cavitation Zone (Bubble Collapse Region)

The cavitation zone is the region inside or immediately after the constriction where cavitation bubbles collapse. This zone experiences highly localized extreme conditions temperatures up to 5000 K, pressures above 1000 atm, and intense hydrodynamic shear. These effects lead to the mechanical disintegration of particles, oxidation of pollutants, generation of hydroxyl radicals, and breakdown of complex organics. It is the primary zone responsible for pollutant degradation and improved biodegradability.

3.5 Methodology

3.5.1 Treated Spent Wash

1. Treated spent wash (TSW) was treated using hydrodynamic cavitation (HC) at pilot scale.
2. TSW was filled into a 200-Lit. cavitation tank.
3. The TSW was directed to the HC system through an SS-316 venturi by opening the suction valve. The recycle valve was fully open, and the discharge valve should be closed.
4. The discharge valve was gradually opened to allow flow through the venturi and regulate the flow to the design value using the recycle valve.
5. The inlet flow rate governs the inlet pressure of TSW. Batch experiments were carried out at inlet flow rates of 3 m³/hr, with samples collected after each pass up to four passes.
6. The flow rate of 3 m³/hr requires 240 to complete one single pass.
7. TSW samples after each pass were collected for further analysis, such as pH, Conductivity, ORP, TS, VS, TDS, TSS, BOD and COD, and BMP.
8. The cavitation unit was switched off after collection of 4 passes, and the pre-treated TSW were disposed off as per industry protocol.

3.5.2 Raw Spent Wash

1. Raw spent wash (RSW) was treated using hydrodynamic cavitation (HC) at pilot scale.
2. RSW were filled into a 200 Lit. cavitation tank.
3. The RSW was directed to the HC system through an SS-316 venturi by opening the suction valve. The recycle valve was fully open, and the discharge valve should be closed.
4. The discharge valve was gradually opened to allow flow through the venturi and regulate the flow to the design value using the recycle valve.
5. The inlet flow rate governs the inlet pressure of RSW. Batch experiments were carried out at inlet flow rates of 3 m³/hr, with samples collected after each pass up to four passes.
6. The flow rate of 3 m³/hr requires 240 seconds to complete one single pass.
7. TSW samples after each pass were collected for further analysis, such as pH, Conductivity, ORP, TS, VS, TDS, TSS, BOD, COD, and BMP.
8. The cavitation unit was switched off after collection of 4 passes, and the pre-treated RSW were disposed off as per industry protocol.

Table 3.5 Flow time calculations of RSW & TSW

Flow of RSW & TSW	3	m ³ /hr
	50.00	L/min
Tank Volume	200	L
Single pass time	240	Sec.
2nd pass time	480	Sec.
3rd pass time	720	Sec.
4th pass time	960	Sec.

Chapter 4

RESULTS AND DISCUSSION

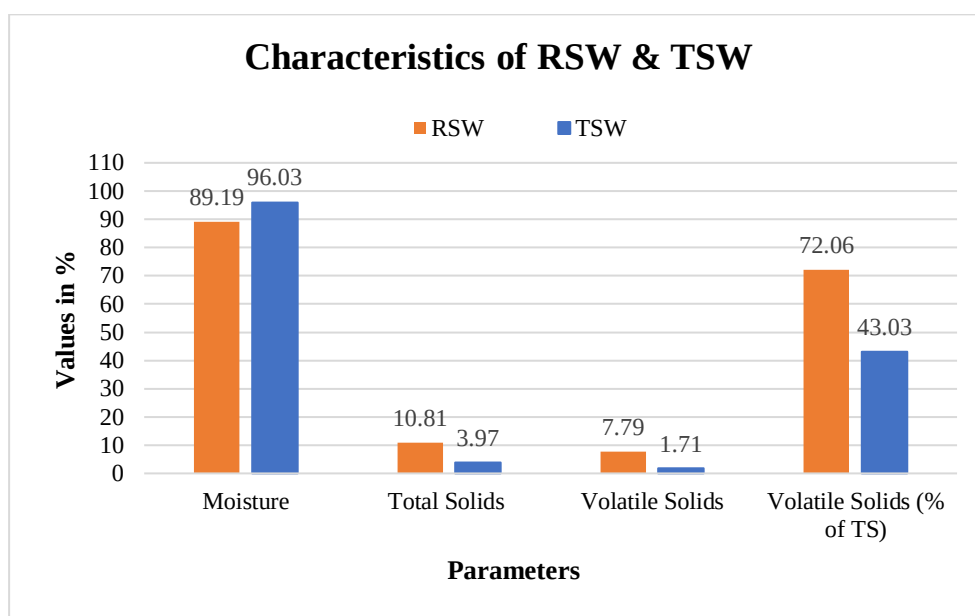
4. Results and Discussions

4.1 Initial characterization of Raw Spent Wash and Treated Spent Wash

The initial phase of this study involved the characterization of Raw Spent Wash (RSW) and Treated Spent Wash (TSW) received from distillery. Table 4.1 (a) presents Initial Characterization for total solids, volatile solids, and moisture content of the RSW and TSW. To improve accuracy and reliability, all samples were analysed in duplicate and average values were taken. Graph 4.1 (b) and (c) present the graphical representation of the initial characteristics of both RSW and TSW.

Table 4.1 (a) Initial Characterization of RSW and TSW

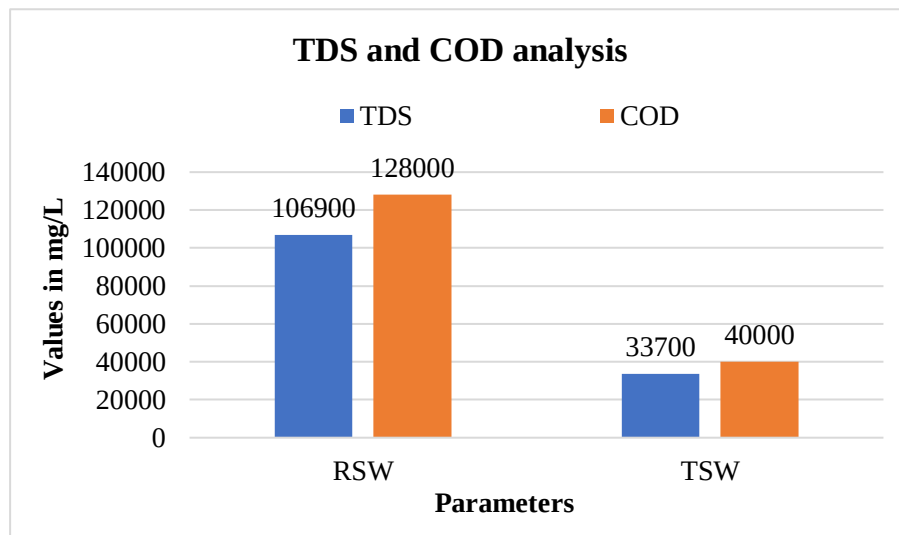
Sr. No.	Parameters	Unit	RSW	TSW
1	Color	-	Brown	Dark brown
2	pH	-	4.37	8.2
3	ORP	mV	65	-250
4	Total Solids (TS)	%	10.81	3.97
5	Volatile Solids (VS)	%	7.79	1.71
6	VS/TS	%	72.06	43.07
7	Total Dissolved Solids (TDS)	mg/L	106900	33700
8	Chemical Oxygen Demand (COD)	mg/L	128000	40000



Graph 4.1 (b) Characteristics of RSW & TSW

The characterization of raw spent wash indicates a high moisture content having value as (89.19%) with moderate total solids (10.81%), the effluent is liquid but still contains a significant solid fraction. The volatile solids (7.79%) represent about 72.06% of the total solids (VS/TS ratio), highlighting a high proportion of biodegradable organic matter. Overall, the high VS/TS ratio shows that raw spent wash contains a large amount of organic matter and is well-suited for anaerobic digestion and biogas production.

The analysis of treated spent wash shows that it is highly diluted, with moisture content around 96% and low total solids (3.97%), indicating that most suspended and dissolved matter has been removed during prior treatment stages. The volatile solids fraction (43% of TS) is considerably lower compared to raw spent wash, which shows that a large portion of easily biodegradable organic matter has already been degraded.



Graph 4.1 (c) Analysis of TDS & COD for RSW & TSW

The TDS value of 106900 mg/L indicates a high concentration of dissolved solids such as organic compounds, salts, and fermentation residues present in the raw spent wash. The COD value of 128000 mg/L reflects high organic load, mainly due to sugars, organic acids, melanoidins, which has high potential for biogas production through anaerobic digestion.

The TDS (33700 mg/L) indicates the presence of dissolved salts and residual organic matter after treatment. The COD value of 40000 mg/L indicates that the treated spent wash still contains residual refractory compounds, representing non-biodegradable

COD (nbCOD). This fraction consists of refractory organic compounds that are resistant to biological degradation.

4.2 Hydrodynamic Cavitation of Raw Spent Wash

Hydrodynamic cavitation of RSW involves passing the effluent through a venturi, which generates microbubbles. The implosive collapse of these bubbles produces extreme localized conditions like high temperatures and pressures, driving the breakdown of recalcitrant organic macromolecules into simpler compounds. This process yields slight reductions in COD after HC treatment with pass 4. In the present study, the hydrodynamic cavitation setup was operated at a constant flow rate of 3 m³/hr across four sequential passes, with each pass requiring approximately 4 minutes (Table 4.2). The fourth pass was selected as the optimal endpoint, as it corresponded to the differential improvement in key parameters (organic fraction and COD). The RSW and RSW pass 4 samples, both before and after HC treatment, were then subjected to BMP analysis to find out enhancement in methane potential.

Table 4.2 Parameters of Raw Spent Wash after HC

TRIAL 1 - RSW (3 m³/hr, 4 min.)					
Parameters	RSW	Pass 1	Pass 2	Pass 3	Pass 4
Color	Brown	Brown	Brown	Brown	Brown
pH	4.37	4.87	4.82	4.91	4.94
ORP (mV)	65	36	54	64	80
Conductivity (µS/cm)	32900	33600	34700	34200	34500
Total Solids (TS) %	9.88	9.54	-	-	10.13
Volatile Solids (VS) %	6.82	6.52	-	-	7.03
VS/TS %	69.03	68.34	-	-	69.39
Chemical Oxygen Demand (COD)	128000	128000	-	-	124000

4.3 Hydrodynamic Cavitation of Treated Spent Wash

Hydrodynamic cavitation applied to treated spent wash, where collapsing microbubbles create reactive conditions. The COD concentration of 40000 mg/L in TSW indicates the presence of residual refractory organic compounds, which contribute to the non-biodegradable COD (nbCOD) fraction. These compounds are resistant to microbial degradation and therefore remain unutilized in biological treatment processes. To convert the nbCOD into biodegradable COD, HC trials were conducted. To evaluate the

improvement in biodegradability after HC treatment, the samples before and after HC were subjected to BMP analysis to determine the enhancement in methane potential.

In the present study, the HC setup was operated at a constant flow rate of 3 m³/hr across four sequential passes, with each pass requiring approximately 4 minutes presents in Table 4.3. The fourth pass was selected as the optimal endpoint, as it corresponded to the differential improvement in COD parameter.

Table 4.3. Parameters of Treated Spent Wash after HC

TRIAL 1 - TSW (3 m³/hr, 4 min.)					
Parameters	TSW	Pass 1	Pass 2	Pass 3	Pass 4
Color	Dark brown	Dark brown	Dark brown	Dark brown	Dark brown
pH	8.20	8.35	8.42	8.44	8.50
ORP (mV)	-250	-325	-60	-38	-20
Conductivity (μS/cm)	37300	35500	36300	36500	36400
Total Solids (TS) %	3.97	-	6.06	-	5.98
Volatile Solids (VS) %	1.71	-	2.76	-	2.72
VS/TS %	43.03	-	45.52	-	45.52
Total Dissolved Solids (TDS) %	3.37	-	3.44	-	3.43
Total Suspended Solids (TSS) %	5.54	-	5.62	-	9.33
Chemical Oxygen Demand (COD)	40000	-	32000	-	28000

4.4 Biochemical Methane Potential (BMP) of Raw Spent Wash & Treated Spent Wash

The Biochemical Methane Potential of RSW and TSW helps to determine methane yield that can be generated from their biodegradable organic fraction under controlled anaerobic conditions. RSW is characterized by high COD, volatile solids and a large proportion of soluble organic matter, making it a highly concentrated and energy-rich substrate for anaerobic digestion. As a result, it generally exhibits higher BMP values. In addition, the high VS/TS ratio in RSW confirms the presence of biodegradable organic matter, which enhances microbial conversion efficiency into methane. However, the presence of inhibitory substances such as phenolic compounds and melanoidins can negatively impact methanogenic microorganisms. These inhibitors may reduce digestion efficiency unless appropriate strategies such as dilution, pH adjustment, nutrient balancing, or pre-treatment methods are applied.

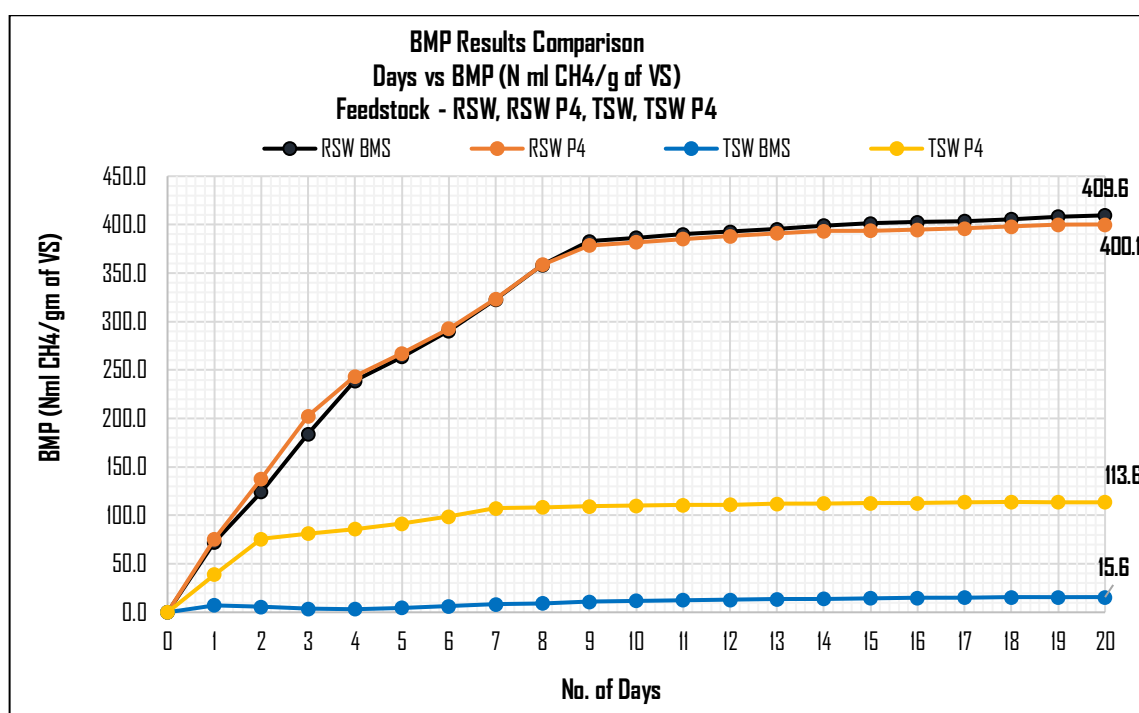
On the other hand, TSW shows comparatively lower BMP values due to partial stabilization and reduction of biodegradable organics during prior treatment processes. The decrease in COD and VS values indicates that a significant fraction of easily degradable matter has already been removed. In this study, BMP analysis was performed on both samples obtained after the 4th pass of hydrodynamic cavitation over a digestion period of 20 days. This duration allows sufficient time to evaluate the ultimate methane yield and assess the biodegradability of both untreated and treated samples. Table 4.4 (a) shows the cumulative methane production over 20 days for RSW, RSW Pass-4, TSW, and TSW Pass-4 samples. Graph 4.4 (b) presents the graphical representation of the BMP values for RSW, RSW Pass-4, TSW, and TSW Pass-4 samples.

Table 4.4. (a) BMP of Raw Spent Wash & Treated Spent Wash before and after HC

No. of Days	BMP of RSW ml CH₄/g VS	BMP of RSW P4 ml CH₄/g VS	BMP of TSW ml CH₄/g VS	BMP of TSW P4 ml CH₄/g VS
1	72.3	75.8	7.1	39.0
2	124.2	137.8	5.5	75.5
3	184.2	202.4	3.6	81.3
4	238.8	243.6	3.2	85.9
5	263.8	267.4	4.7	91.7
6	290.5	292.7	6.3	98.8
7	322.9	323.6	8.2	107.4
8	358.5	358.7	9.4	108.2
9	383.0	378.6	11.0	109.2
10	386.5	381.7	11.9	110.2
11	390.0	385.3	12.6	110.8
12	392.9	388.1	12.9	111.0
13	395.6	391.1	13.5	111.9
14	399.0	393.4	13.9	112.4
15	401.5	393.8	14.5	112.6
16	402.8	394.8	14.9	112.8
17	403.7	396.2	15.2	113.5
18	405.6	398.2	15.4	113.5
19	408.2	400.1	15.6	113.6
20	409.6	400.1	15.6	113.6

The above BMP data indicate that HC treatment had a negligible effect on RSW, as the methane yield values remained almost unchanged and were slightly lower than those of

untreated RSW. The BMP values for RSW and RSW Pass-4 were 410 and 400 NmL CH₄/g VS, respectively. However, HC treatment significantly improved the biodegradability of TSW, as reflected by the considerable increase in methane yield in TSW Pass-4, as BMP values increased from 15.6 to 113.6 NmLCH₄/ gm VS.



Note: P4 represents sample after HC Pass 4

Graph 4.4 (b) BMP Results Comparison of RSW and TSW with HC Pass 4

Raw Spent Wash (RSW) showed the highest methane production, reaching approximately 410 NmL CH₄/g VS, indicating the presence of a high fraction of biodegradable organic matter. Both RSW and RSW P4 methane production curves exhibited a rapid increase up to day 8–9, followed by a plateau phase, showing that the readily degradable substrate was quickly consumed during the early digestion period. Although a slight increase in BMP was initially observed in the RSW pass 4 (after HC) sample compared to untreated RSW, a decrease was observed after the 9th day until the 20th day of the digestion period. This reduction may be attributed to excessive oxidation of organic matter during HC treatment, which could have reduced the methane production potential.

Therefore, HC treatment up to pass 4 did not significantly improve the methane potential of RSW, confirming that pretreatment offers minimal benefit for substrates that are already highly biodegradable.

In contrast, treated spent wash (TSW) showed a low BMP of about 15.6 NmL CH₄/g VS, indicating that most of the biodegradable organic matter had already been removed during prior treatment, leaving mainly refractory compounds. However, after hydrodynamic cavitation (HC) pretreatment (TSW pass 4), the BMP increased significantly to 113.6 NmL CH₄/g VS, corresponding to an approximately 7.2-times increase in methane potential. This indicates that HC improved the biodegradability of the residual organic matter by converting a portion of non-biodegradable COD into biodegradable fractions, thereby enhancing methane production.

Therefore, hydrodynamic cavitation can be considered a useful pretreatment strategy for enhancing methane recovery from treated distillery effluents.

Chapter 5

Conclusion and Future Scope

5. Conclusion and Future Scope

5.1 Conclusion

1. The characterization results indicate that Raw Spent Wash (RSW) contains a high amount of organic matter. The volatile solids fraction VS/TS as 72.06% shows the presence of a large biodegradable organic fraction, making it suitable for anaerobic digestion and biogas production.
2. Treated Spent Wash (TSW) shows a significantly low VS/TS ratio as 43.03% reflecting prior anaerobic digestion treatment and depletion of readily degradable organics.
3. Hydrodynamic cavitation (HC) pre-treatment enhances organic solubilization through cavitation-induced phenomena, implosive micro-jetting and hydroxyl radical ($\cdot\text{OH}$) facilitating degradation of polymeric compounds of recalcitrant macromolecules (melanoidins, phenolics).
4. The BMP values for RSW and RSW Pass-4 were 410 and 400 NmL $\text{CH}_4/\text{g VS}$, respectively. This data indicates that HC treatment had a negligible effect on RSW, as the methane yield values remained almost unchanged and were slightly lower than those of untreated RSW.
5. The reduction in BMP value of RSW after HC may be attributed to excessive oxidation of organic matter during treatment, which could have reduced the methane production potential.
6. However, HC treatment significantly improved the biodegradability of TSW, as reflected by the considerable increase in methane yield in TSW Pass-4, as BMP values increased from 15.6 to 113.6 Nml $\text{CH}_4/\text{gm VS}$.
7. In case of TSW, HC improved the biodegradability of the residual organic matter by converting a portion of non-biodegradable COD into biodegradable fractions, thereby enhancing methane production corresponding to an approximately 7.2-times increase in methane potential.
8. In both RSW and TSW, no colour reduction was observed after Hydrodynamic cavitation treatment.

5.2 Future Scope

1. Exploration of hybrid HC–AOP systems (HC + H₂O₂, O₃, Fenton oxidation) for improved degradation of recalcitrant compounds such as melanoidins and phenolics.
2. Exploration of hydrodynamic cavitation (HC) at different number of passes and flow rates and its effect on BMP.
3. Implementation of co-digestion strategies involving TSW with high-strength substrates (e.g., press mud or manure) to improve the C/N ratio and methane yield.
4. Evaluation of pilot- and industrial-scale feasibility, including techno-economic analysis.
5. Study the effect of different pH conditions (lime addition) during cavitation and digestion to find the most favourable condition to improve methane yield or BMP.
6. Study the effect of HC on color removal, especially targeting measurement of melanoidins in spent wash.
7. Expanding the use of hydrodynamic cavitation (HC) to other high-strength industrial effluents to improve treatment efficiency and promote wider use of this technology.

Chapter 6
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6. REFERENCES

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2025-2026

YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING NAGPUR
(an autonomous Institution Affiliated to Rashtrasant Tukdoji Maharaj Nagpur University)
DEPARTMENT OF CIVIL ENGINEERING
“Study on Pre-treatment of Cane molasses based Distillery Raw Spent Wash using
Hydrodynamic Cavitation for Enhanced Biogas Production”
NAME OF STUDENT: VEDHAS M. DORLIKAR
NAME OF THE GUIDE: Dr HARSHAL M. WARADE.



ABSTRACT

This study investigates hydrodynamic cavitation (HC) as a pre-treatment method for distillery spent wash to enhance biodegradability and methane production. HC breaks complex organic compounds into simpler biodegradable forms, improving BOD/COD ratio, and BMP yield. The results confirm that HC is an energy-efficient, eco-friendly, and scalable technology for sustainable wastewater treatment and biogas generation.

INTRODUCTION

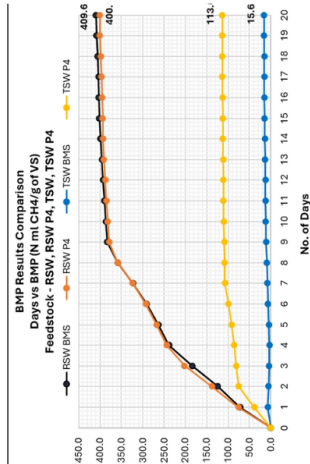
- Distillery spent wash is a high-strength wastewater generated during ethanol production from molasses.
- It contains high COD, BOD, dark colour, and refractory organic compounds.
- Direct discharge of spent wash causes serious environmental pollution.
- Conventional biological treatment alone is often insufficient for complete treatment.
- Hydrodynamic cavitation (HC) breaks complex pollutants into simpler biodegradable compounds.
- HC improves anaerobic digestion and enhances methane production.
- It also reduces toxicity and increases wastewater biodegradability.

Therefore, HC is a sustainable and energy-efficient solution for distillery wastewater management.

METHODOLOGY

- Raw spent wash (RSW) and treated wastewater (TSW) samples were collected from a molasses-based distillery.
- Initial physicochemical characteristics such as pH, COD, TS, VS, and colour were analyzed.
- Samples were subjected to hydrodynamic cavitation treatment at a flow rate of 3 ml/min operating at 1000 rpm.
- Different cavitation passes were applied to study treatment efficiency, in which pass 4 was optimum.
- After HC treatment, changes in wastewater properties were measured.
- Biochemical Methane Potential (BMP) tests were conducted using the AMPTS system.
- Methane yield of untreated and HC-treated samples was compared.
- The effectiveness of HC was evaluated based on pollutant reduction and enhancement in biogas production.

RESULTS



TRIAL - TSW & RSW (3 m ³ /hr, 4 min.)				
Parameter	TSW	Pass 4	RSW	Pass 4
Color	Dark	Dark	Brown	Brown
pH	8.2	8.5	4.37	4.94
ORP (mV)	-250	-20	65	80
Conductivity	37300	36400	32900	34500
Total Solids (TS)	3.97	5.98	9.88	10.13
Volatile Solids	1.71	2.72	6.82	7.03
VS/TS %	43.03	45.52	69.03	69.39
Total Dissolved	3.37	3.43	-	-
Total Suspended	5.54	9.33	-	-
Chemical Oxygen Demand	40000	28000	128000	124000

CONCLUSION & FUTURE SCOPE

CONCLUSION

- Hydrodynamic cavitation (HC) significantly reduced the color and ORP of distillery spent wash, indicating effective treatment of refractory organics.
- The reduction in BMP value of RSW after HC may be due to the breakdown of complex organic compounds.
- However, HC treatment significantly improved the biodegradability of TSW, as reflected by the considerable increase in VS/TS ratio from 43.03% to 45.52%.

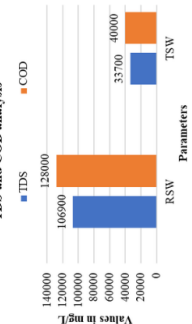
FUTURE SCOPE

- Optimization of pressure, flow rate, and number of cavitation passes for industrial applications.
- Integration of HC with anaerobic digesters for continuous treatment of HC with enhanced oxidation or biological treatment.
- Techno-economic and energy consumption analysis at large scale.
- Application of HC for other industrial wastewaters and sludge treatment.

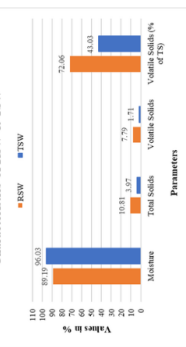
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TDS and COD analysis



Characteristics of RSW & TSW



Sr. No.	Parameters	Unit	RSW	TSW
1	Color	-	Brown	Dark brown
2	pH	-	4.37	8.2
3	ORP	mV	65	-250
4	Total Solids (TS)	%	10.81	3.97
5	Volatile Solids (VS)	%	7.78	1.71
6	VS/TS	%	72.06	43.07
7	Total Dissolved Solids (TDS)	mg/L	106900	33700
8	Chemical Oxygen Demand (COD)	mg/L	128000	40000

ANNEXURE



Microsoft CMT <noreply@msr-cmt.org>

to me ▾

Tue, Jan 20, 4:54 PM



Dear VEDHAS MILIND DORLIKAR

Paper ID / Submission ID : 145

Title : Hydrodynamic Cavitation: Principles, Applications, and Challenges in Wastewater Treatment

Greeting from ICSIETS2026.

We are pleased to inform you that your paper has been accepted for the Oral Presentation as a full paper for International conference on Sustainability and Innovation in Engineering, Technology and Applied Sciences, NAGPUR, INDIA scheduled from 26th February to 27th February 2026.

All accepted and presented papers will be submitted to GRENZE Scientific Society for Scopus Indexed international conference proceedings publication.

Step1- Proceed for Registration Process and Camera-Ready Paper Submission within 03 days from the receipt of this Email.

The registration details are available at following Website



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